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TECHNICAL PAPER

Validation and real-time application of a novel order-separated feedforward controller with PI feedback for electrohydraulic actuation of linear motion

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Abstract
For electrohydraulic systems with double-acting cylinders, existing feedforward controllers require close matching of the flow-area ratio across the piston with that of the peripheral lengths of the corresponding metering valve ports cut along its sleeve. Existing feedforward models do not consider variable pump displacement and, in case of a single-rod cylinder, ignore the effect of oil compressibility as well. Moreover, the effect of either flow leakage in valves or cylinder friction was neglected. A novel order-separated feedforward, or OSFF, model has been proposed here to include all these issues under a generally acceptable assumption for the piston motion-induced flow to dominate over other flows. Also, a conventional feedforward, or CFF, model has been constructed, for which the matching constraint could not be relaxed. Offline GA-based minimization of the deviation between an experimental result and the corresponding simulations has been used to estimate the parameters of both the feedforward models. By coupling a PI feedback in real time to form OSFFPI and CFFPI controllers, the effects of the assumptions and approximations in the feedforward parts has been compensated. Using a proposed extension of Routh stability criteria for the linearized time-variant error dynamics, the PI gains have been selected. The response of the OSFFPI controller has exhibited not only better tracking up to 4 Hz sinusoidal-frequency demand, but also lower control effort over each of the CFFPI controller, an existing nonlinear adaptive sliding-mode controller and the simplest but the worst PI-only controller. While the comparison with the CFFPI scheme has ascertained the acceptability of the more generalized OSFFPI controller, the superior performance than an existing controller is a notable contribution. Compared to the fixed-displacement pump mode, the experiments under the two proposed schemes have shown higher energy efficiency of the OSFFPI controller except for very low-velocity motion demand.

Keywords Electrohydraulic control · Feedforward modelling · Identification · Real-time experiments

1 Introduction

Electrohydraulic systems are more compact and capable of negotiating higher external force or torque than their electrical counterparts that are admittedly faster, but less fault tolerant [1]. Electromechanical actuators have in-between speed of response, but possess poor back-drivability [2] implying high proneness to failure under sudden load reversal. Based on their strengths, electrohydraulic systems are widely used in agricultural [3], automobile [4], aerospace [5–7], construction [8, 9], mining [10] and manufacturing [11, 12] industries and for measuring certain mechanical properties of materials [13, 14]. Their good back-drivability allows to handle even negative loading [15]. However, nonlinear characteristics [1] arising from pressure–discharge relation at the metering valve ports, flow leakage in the valves, cylinder friction and oil compressibility make precision tracking control quite challenging. Systems with proportional valves and industry-grade cylinders are cheaper yet rugged, but involve deadband [16, 17] and static friction [18] nonlinearities that are negligible in systems with servo-proportional or servo valves and low-friction cylinders. Model-free controllers were successfully used in electrohydraulic

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systems [7, 11, 17, 19]. But, the absence of a physical model neither ensures their optimal performance nor allows performing any stability analysis. In this regard, model-based controllers [3–6, 8–10, 12, 13, 16, 18] are better.

A feedforward controller [20–26] requires a differentially flat model for using the output or the demand and the corresponding derivatives for estimating the valve command, which compensates the known nonlinear effects and disturbances involving much lower uncertainties. Such models were exploited in systems [25–29] with a fixed-displacement pump feeding a double-acting cylinder through a valve with close matching of the ratios of the flow areas across the piston and the curved lengths of the corresponding metering valve ports cut on the sleeve. However, the estimation became numerically very stiff in case of single-rod cylinders, with oil compressibility assuming significance at high working pressure [27]. In a system with a double-rod cylinder, the compensation corresponding to the oil compressibility-induced force that closely met the external demand was estimated from a feedforward model and that for the system friction in a separate impedance-matching loop [28]. A feedforward controller was developed for a system with a single-rod cylinder fed by a variable-displacement pump under the assumptions of negligible oil compressibility and cylinder friction [20]. The effect of flow leakage in the valves and cylinders was neglected in all these models [20]. In comparison with these feedforward models developed by invoking different sets of restrictions, a more generalized feedforward model would indeed be more valuable.

In the past, a number of adaptive controllers [5, 30–33] were developed that were free from the limitations of the existing feedforward models. A feedforward model is routinely combined with a feedback controller, where the former compensates the known effects that in most cases dominate over the uncertainties and approximations tackled by the latter part [2, 25–27]. Such handling of the dominating part on the basis of sound physical feedforward model is more likely to be optimal than the adaptive controllers. The real-time use of a feedforward controller calls for estimation of a number of feedforward parameters, for which standard identification processes exist [5, 8, 23, 24]. However, the setting of a number of parameters required for real-time execution of an adaptive controller is heavily dependent on the users' skill.

Under the status outlined above, the task has been set here as feedforward, or FF, modelling in two alternative ways along with their real-time implementation with a PI feedback in a simple electrohydraulic set-up for linear motion actuation. Section 2 contains the system modelling with reference to an existing set-up, in which all the experiments have been conducted. Since the feedback excitation has been included only to take care of the modelling simplifications, it would be dominated by the feedforward excitation in the overall sense. The contributing aspect of both the models is their broader applicability over the existing models through combined inclusion of the effects of variable pump pressure, oil compressibility, cylinder friction and flow leakages in the valve and the cylinder.

In Sects. 3–5, the feedforward modelling and controller formulations are described. By using available basic models or experimental findings for each individual aspect, one model leads to a consolidated estimation of the feedforward voltage excitation in a conventional manner. The consequent conventional controller, hereafter referred as CFFPI, has been used to test the acceptability of the novel order-separated feedforward model giving rise to another controller, named here as OSFFPI. An acceptability test becomes necessary for using a new leakage model in the novel approach along with splitting the feedforward estimation by exploiting the naturally emerging higher-order effect that induces the piston motion in comparison with each corresponding to the flow leakage and oil compressibility. Such an approach leads to a more generalized model not necessitating to hold the ratio matching requirement that could not be relaxed in the conventional scheme. Sections 6 and 7 detail the identification procedure and estimation of the feedforward parameters, respectively. An existing GA-based optimization code [34] has been used offline to accomplish the identification that results in the minimum deviation between an experimental variation and a number of predictions corresponding to different parameter sets for each of the two proposed models.

A notable contribution of the study involves extending the stability criteria due to Routh–Hurwitz for the linearized error dynamics equation with time-varying coefficients and using the outcome of the extended procedure to select the gains of the PI feedback controller. In Sect. 8, the procedure has been explained, and Sect. 9 contains the results that provide the choice of P and I gain for the CFFPI and OSFFPI controllers. The performance of these controllers is compared in Sect. 10 against an existing adaptive sliding mode controller combined with a nonlinear integral control [30]. In order to use the existing code for validation, the pump has been operated in the fixed-displacement mode so as to keep the delivery pressure constant at the relief valve setting. On the other hand, the acceptability tests have been carried out with the pump working in the variable-displacement mode. A comparison of energy-saving has also been given, as achieved under the two proposed controllers by operating the pump in the same mode with respect to the fixed-displacement mode pump operation. The conclusions are drawn in Sect. 11 along with outlining the scope of future studies.

2 System description and modelling

All the experiments for the study reported here have been carried out in an existing set-up described next with reference to Figs. 1a and b along with Tables 1 and 2. Figure 1a

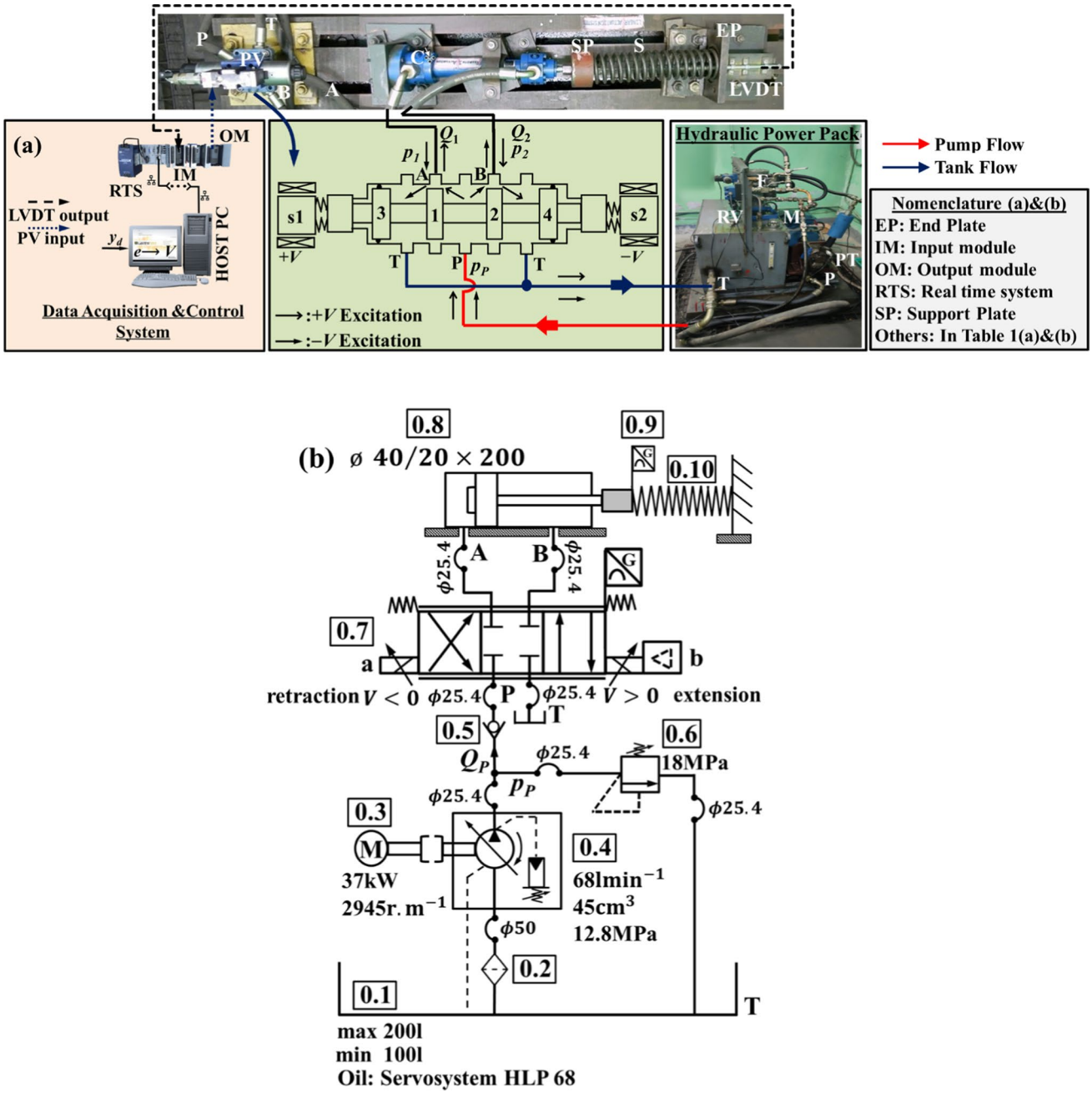


Fig. 1 Electrohydraulically actuated linear-motion set-up in **a** photographic view and **b** circuit diagram

Table 1 Component specifications of real-time system

Component description	Make	Component specification
Input Module **IM**	NI systems	NI 9215 (4 Channel; ± 10 V A/D, 16-bit simultaneous analog input)
Output module **OM**	NI systems	NI 9263 (± 10 V D/A,16-bit simultaneous analog output)
Real-time system **RTS**	NI systems	1 GHz, 32-bit NI-cRIO 9081, Sampling rate-100 KS/s

and b, respectively, depicts the photographic view and the circuit diagram. The diagram conforms to ISO 1219: 2–2012 along with a pump symbol specific for the set-up [35]. Under active command signalling, a proportional valve **0.7** [36] controls the oil flow from a variable-displacement pump **0.4** [37] to drive a piston within the bore of an industry-grade

Table 2 Component specifications of electrohydraulic set-up

Code number	Component description	Make	Component specification
0.1	Reservoir (**T**)	Local	Dim. – 1.0 m × 0.6 m × 0.5 m; capacity – 200×10^{-3} m^3
0.2	Filter (**F**)	Stauff	SF045B/TB/B; Maximum working pressure 4.2×10^7 Pa
0.3	Motor (**M**)	ABB	3GBA201044-ADEIN; 3-phase; 2 pole: 37 kW; operating speed 1250 rpm with VFD
0.4	Pressure compensated variable displacement pump, (**P**)	Rexroth [37]	A10VSO 45 DR/31R-PPA12N00; operating discharge 8.3×10^{-4} m^3 s^{-1} at motor speed 1250 rpm with VFD
0.5	Inline check valve	Yuken	CIT-06–2080;
0.6	Relief valve (**RV**)	Yuken	EBG-03-H-11; 2.05×10^7 Pa
0.7	Proportional directional valve (**PV**)	Rexroth [36]	4WREE 10 E50-23/G24K31/A1V
0.8	Double-acting single-rod cylinder (**C**)	Rexroth [38]	CD250D40/20-200A1X/01-CGDHT; full stroke: 200 mm; bore diameter: 40 mm; rod diameter: 20 mm; piston mass 2.5 kg
0.9	Displacement sensor (**LVDT**)	HBM	Plunger type; 0–0.200 m range with 0.5–10 V analog output; Nominal Sensitivity: 80 mV/V
0.10	Spring (**S**)	Local	Stiffness 20 kNm^{-1}

double-acting single-rod cylinder **0.8** [38] fed by the pump **0.4** driven by a constant-speed motor **0.3**.

The pump could be driven in the fixed-displacement mode to keep the delivery pressure p_p constant at the setting of the relief valve **0.6** by always allowing some oil draining back to the reservoir **0.1**, also referred here as tank. When the pump operates in the variable-displacement mode, both the delivery pressure p_p and discharge Q_p keep on varying. The cut-in pressure p_{ci} corresponds to nearly the highest possible discharge Q_{p0}. In order to capture the steep drop of pump discharge to zero with increase in the delivery pressure beyond the cut-in value noted in the experiment [37], the pressure–discharge relation of the pump working in the variable-displacement mode is expressed in terms of a characteristic flow Q_c as

$$p_p \equiv f_1(Q_p) = p_{ci}[2 + \coth\{(Q_p - Q_{p0})/Q_c\}]. \tag{1}$$

Ports **P**, **T**, **A** and **B** in the **0.7** in Fig. 1b are meant for communicating the valve flow with the pump, the tank and the cap and rod-end sides of the cylinder, respectively. A filter **0.2** is meant to remove small particles so as to keep the hydraulic oil circulating between the pump and the tank clean. A check valve **0.5** in the system does not allow any reverse flow through the pump. The cylinder **0.8** is clamped on a horizontal test bed with its axis collinear for the piston rod, a compression spring **0.10** acting as a force generator in the set-up and the rod of a linear variable differential transformer, or **0.9** that measures the actual piston displacement y.

The piston rod is screwed at the left side of a cup that could move through a hole in an end plate **EP**. Within the hollow of the cup, the spring and its guide rod are held unfastened and a support plate **SP** is provided at the other end of the spring. In order to ensure the movement of the piston, its rod, the guide rod and the **LVDT** rod as a single unit leading to a change of the compressed length of the spring, a central hole is drawn in the **SP** to pass the guide rod of the spring through it and connect it with the **LVDT** rod. An input module **IM** routes the **LVDT** signal to a real-time system **RTS** that runs a controller and estimates an excitation voltage V within $\pm V_{\max}$ with $V_{\max}$ equal to 10 V, based on the demand y_d for the piston displacement set through a **Host PC**. The excitation is directed by an output module **OM** to the appropriate solenoid **s1** or **s2** of the **0.7** setting flow through it from the pump to one chamber of the cylinder and directs flow from the other chamber back to the tank. Under no excitation, the valve keeps the cylinder disconnected from both the pump and the tank. Even at the full excitation, substantial pressure drop occurs for sustaining the orifice-like usually sub-mm axial opening of metering ports **A** and **B** of the valve. Consequent pressures p_1 and p_2 at the cap and rod end chambers generate a thrust to push the piston against a friction force F_f and spring force ky exerted by the spring of stiffness k. If the total moving mass of the assembly is m, the dynamics of the piston emerges as

$$\dot{y} = v \text{ and } \dot{v} = \ddot{y} = \left(p_1 A_1 - p_2 A_2 - ky - F_f\right)/m, \tag{2}$$

where $A_1 = \pi d_1^2/4$ and

$$A_2 = \pi(d_1^2 - d_2^2)/4, \tag{3}$$

y, $\dot{y} = v$ and $\ddot{y} = \dot{v}$ are the displacement, velocity and acceleration of the piston, respectively, and d_1 and d_2 are the cylinder bore diameter and piston rod diameter, respectively. In the catalogues [36, 38], the valve and cylinder masses are specified, respectively, as 6.5 kg and 2.5 kg. As apparent from the valve schematic in Fig. 1a, the spool strokes within a sleeve that in turn is housed inside a much bigger body containing all the flow passages. Also, a pair of solenoids

and the associated electronics imply the mass of the tiny spool to be much lower than actuated mass together with the piston. Hence, taking cue from some existing studies [20, 26], the spool dynamics has been neglected with respect to that of the piston.

The friction force [34, 39] F_f arises between the piston seals and the cylinder bore during the motion of the piston or its onset. A relatively simple friction model that conforms to the development of the targeted feedforward model involving static friction F_s, Coulomb friction F_c, Stribeck velocity v_s and viscous or dynamic friction coefficient α_v is expressed as

$$F_f = \{F_c + (F_s - F_c)\exp\{-(\dot{y}/v_s)^2\}\mathrm{sgn}(\dot{y}) + \alpha_v\dot{y}. \tag{4}$$

Even a further simpler version of the form

$$F_f = \{F_c\}\mathrm{sgn}(\dot{y}) + \alpha_v\dot{y} \tag{5}$$

is also quite popular for realizing a feedforward model executable in real time.

Oil leakage takes place through the radial clearance between the spool land and the sleeve at the valve ports **A** and **B**. Two different models of the leakage flow have been proposed in the present work. The first model is an improvisation corresponding to an existing experimental result [40], which indicates the valve leakage to be the maximum at the neutral spool position and decays through a bell-shaped variation with spool displacement away from the neutral. By using l for leakage as subscript or superscript, the model is formulated as

$$Q_l = Q_1^l + Q_2^l = 2Q_{l0}[\exp(-\alpha_l\{(1 - p_1/p_2)/(1 + p_1/p_2)\}^2] + Q_{l1}, \tag{6}$$

along with

$$Q_1^l = \sqrt{p_1}Q_l/\left(\sqrt{p_1} + \sqrt{p_2}\right) \text{ and } Q_2^l = \sqrt{p_2}Q_l/\left(\sqrt{p_1} + \sqrt{p_2}\right), \tag{7}$$

where Ports **A** and **B** in Fig. 1, respectively, involve subscripts 1 and 2, and leakage parameters are Q_{l0}, Q_{l1} and α_l. This model envisages the valve leakage to arise under the forcing by the pressure in each cylinder chamber through the orifice-like radial leakage paths in the form of annular sectors in the bore of the valve sleeve that return the flow to the open tank taken to be at zero gauge pressure.

In the second approach, the leakage at Ports **A** and **B** is modelled as orifice flows through the constant-area radial clearance at the side of the spool land that moves under the sleeve. In the central block of the valve symbol of Fig. 1b with all the ports disconnected, the metering lands at Ports **A** and **B** remain at the neutral position. At this position, the higher width of a spool land than the width of the corresponding port cut in a proportional valve with deadband, constitutes two short radial clearances ending at the right and left edges of each these lands. The spool movement away from the neutral creates an opening on one side through a preceding stage of shortening of the length of the radial clearance. Of course, the shortening of the length ends as the moving circular edge of the spool land comes just under the edges of the sleeve cuts, each of which is a circular arc.

In the valve schematic of Fig. 1, two orifice openings are apparent in the vicinity of each metering land. On the axially opposite side of the same land, the radial clearance keeps on getting longer throughout the spool movement without any increase in the flow area across the clearance. These clearances are responsible for the oil leakage. In comparison with the pressure drops in all these orifices, all other pressure losses in a continuous flow path remain negligible. In terms of usual orifice discharge relation proportional to square root of pressure drop, a leakage model is proposed as

$$Q_l = Q_1^l + Q_2^l : Q_1^l = c_{l1}\sqrt{p_1} \text{ and } Q_2^l = c_{l2}\sqrt{p_p - p_2} \text{ for } \dot{y} \geq 0, \tag{8}$$

and

$$Q_l = Q_1^l + Q_2^l : Q_1^l = c_{l1}\sqrt{p_p - p_1} \text{ and } Q_2^l = c_{l2}\sqrt{p_2} \text{ for } \dot{y} < 0, \tag{9}$$

where c_{l1} and c_{l2} are the leakage constants at Ports **A** and **B**, respectively.

The oil flow through the orifices, getting opened under the voltage excitation to one of the valve solenoids, induces a pressure dynamics in the connected cylinder chambers due to accumulation and depletion of oil in the two chambers. Eventually, the pressure difference developing across the piston sets it in motion. Taking bulk modulus of the hydraulic fluid as β, the equations for the cap end and rod end pressure dynamics are written as

$$\dot{p}_1 = \beta(Q_1 - A_1 v)/(V_{01} + A_1 y), \tag{10}$$

and

$$\dot{p}_2 = \beta(A_2 v - Q_2)/(V_{02} - A_2 y), \tag{11}$$

where V_{01} and V_{02} are the oil volumes, respectively, at the cap and rod end sides of **C** with the piston most retracted and Q_1 and Q_2 are the discharges, respectively, through ports **A** and **B** of **0.7**. Each of the discharges through the metered ports has two parts. The first part is induced by the piston motion and is proportional to the voltage V, valve coefficient c_{vi} and the square root of the pressure drop across the port. From this part, a second part Q_i^l due to the leakage is subtracted to obtain the net flow Q_i entering or leaving the corresponding cylinder chamber. Hence, the flow-pressure model for the metered ports is expressed as

$$Q_1 = c_{v1}V\sqrt{p_p - p_1} - Q_1^l \text{ for } \dot{y} \geq 0, \tag{12}$$

or

$$Q_1 = c_{v1} V \sqrt{p_1} - Q_1^l \text{ for } \dot{y} < 0, \tag{13}$$

$$Q_2 = c_{v2} V \sqrt{p_2} - Q_2^l \text{ for } \dot{y} \geq 0, \tag{14}$$

or

$$Q_2 = c_{v2} V \sqrt{p_p - p_2} - Q_2^l \text{ for } \dot{y} < 0. \tag{15}$$

The different subscripts 1 and 2 have been used for the coefficients, in view of different lengths of the peripheral cuts in the sleeves communicating flow, with the larger cap-end area and smaller rod-end area of the cylinder chambers, respectively. At Port **P** of the valve shown in Fig. 1, the pump flow is received by the valve. This is routed through Ports **A** and **B**, one of which, in the active state of a solenoid, remains open and the other provides a leakage path. The solenoid connections in the proportional valve have been set in a manner so that a positive and negative excitation cause extension and retraction of the piston, respectively. Hence, the pump flow through the proportional valve is expressed as

$$Q_p \equiv f_2(V, p_p, p_1, p_2) = V c_{v1} \sqrt{p_p - p_1} + Q_2^l \text{ for } \dot{y} \geq 0, \tag{16}$$

or

$$Q_p \equiv f_2(V, p_p, p_1, p_2) = V c_{v2} \sqrt{p_p - p_2} + Q_1^l \text{ for } \dot{y} < 0. \tag{17}$$

A valve-cylinder pairing of the form

$$= c_{v2}/c_{v1} = c_{l2}/c_{l1} = A_2/A_1 \text{ and } \Omega = c_{v2}/(\lambda c_{v1}), \tag{18}$$

makes the pressure drops at the metered ports nearly equal for the domination of the motion-inducing discharge proportional to the piston velocity over the leakage expressed by (12–15) and the compressibility effect captured by (10) and (11). Under these assumptions, (12–15) yield

$$p_p = p_1 + p_2, \tag{19}$$

$$p_2 = p_p/\Omega^2 - p_1/\Omega^2 \text{ for } \dot{y} \geq 0, \tag{20}$$

and

$$p_2 = -p_p - p_1/\Omega^2 \text{ for } \dot{y} < 0. \tag{21}$$

Equal pressure drop at the paired metering ports of a proportional valve implies matched response of pressure change in these ports during any transient. Unequal drop would have led to different transient durations induced by the variation of pressures in the flow through the two ports. Consequently, the overall transient time would be longer. A transient process could set in through change of the external loading on the piston or the delivery pressure of the pump working in variable-displacement mode.

As evident from (16) and (17), the pump flow is governed by the pressures in the two cylinder chambers corresponding to the external spring loading evident in Fig. 1a for the system under study. Under the spring loading and use of (4) and (5) as friction model, respectively, for the conventional and order-separated models, the use of (20) and (21) in (2) yields the expressions for pressures in the two cylinder chambers as

$$p_1 \equiv f_3(p_p, \ddot{y}, \dot{y}, y) = \left\{ (p_p/\Omega^2)\lambda A_1 + m\ddot{y} + ky + F_f \right\} / \left\{ A_1 (1 + \lambda/\Omega^2) \right\} \text{for} \dot{y} \geq 0, \tag{22}$$

or

$$\left\{ p_p \lambda A_1 + m\ddot{y} + ky - F_f \right\} / \left\{ A_1 (1 + \lambda/\Omega^2) \right\} \text{ for } \dot{y} < 0, \tag{23}$$

and

$$p_2 \equiv f_4(p_p, \ddot{y}, \dot{y}, y) = \left\{ (p_p/\Omega^2) A_1 - (m\ddot{y} + ky + F_f)/\Omega^2 \right\} / \left\{ A_1 (1 + \lambda/\Omega^2) \right\} \text{for} \dot{y} \geq 0, \tag{24}$$

or

$$\left\{ p_p A_1 - (m\ddot{y} + ky - F_f)/\Omega^2 \right\} / \left\{ A_1 (1 + \lambda/\Omega^2) \right\} \text{for} \dot{y} < 0. \tag{25}$$

The expressions for the pressure transients in the cylinder chambers are rewritten by using (12–15) in (10) and (11), respectively, as

$$\dot{p}_1|_{\dot{y} \geq 0} = \beta [c_{v1} V \sqrt{p_p - p_1} - Q_1^l - A_1 \dot{y}] / (V_{01} + A_1 y), \tag{26}$$

or

$$\dot{p}_1|_{\dot{y} < 0} = \beta [c_{v1} V \sqrt{p_1} - Q_1^l - A_1 \dot{y}] / (V_{01} + A_1 y), \tag{27}$$

and

$$\dot{p}_2|_{\dot{y} \geq 0} = \beta [\lambda A_1 \dot{y} - c_{v2} V \sqrt{p_2} + Q_2^l] / (V_{02} - \lambda A_1 y), \tag{28}$$

or

$$\dot{p}_2|_{\dot{y} < 0} = \beta [\lambda A_1 \dot{y} - c_{v2} V \sqrt{p_p - p_2}\} + Q_2^l\}] / (V_{02} - \lambda A_1 y). \tag{29}$$

Finally, by differentiating (2) and using (26–29), a third-order motion dynamics is obtained as

$$\dddot{y} \equiv f_5\left(\ddot{y}, \dot{y}, y, V, p_p, p_1, p_2\right) = \{\beta A_1\left(c_{v1}V\sqrt{p_p - p_1} - Q_1^l - A_1\dot{y}\right)/\left(V_{01} + A_1 y\right) - \beta\lambda A_1(\lambda A_1\dot{y} - c_{v2}V\sqrt{p_2} + Q_2^l)/\left(V_{02} - \lambda A_1 y\right) - k\dot{y} - \alpha_v\ddot{y}\}/m \text{ for } \dot{y} \geq 0, \tag{30}$$

or

$$\{\beta A_1\left(c_{v1}V\sqrt{p_1} - Q_1^l - A_1\dot{y}\right)/\left(V_{01} + A_1 y\right) - \beta\lambda A_1(\lambda A_1\dot{y} - c_{v2}V\sqrt{p_p - p_2} + Q_2^l)/\left(V_{02} - \lambda A_1 y\right) - k\dot{y} - \alpha_v\ddot{y}\}/m \text{ for } \dot{y} < 0. \tag{31}$$

It is evident from (30) and (31) that the piston motion is set in by the high-pressure flow delivered by the pump through the proportional valve to the cylinder and back to the open tank. The variation of pump pressure in a variable-displacement pump in this set-up gives rise to a system transient that do not arise in case of a fixed-displacement pump. A feedforward compensation is considered in controller formulation discussed next.

3 Overall feedforward formulation

An FFPI position controller has been developed for a system with a pressure-compensated variable-displacement pump and low-cost as well as rugged pair of a single-rod double-acting cylinder and a proportional valve. As depicted in Fig. 2a, the controller targets minimizing the error defined as

$$e = y_d - y. \tag{32}$$

The control voltages for the FFPI controllers are evaluated as

$$V = V_f + v_b \,:\, v_b \ll V_f, \tag{33}$$

where V_f and v_b are the feedforward and feedback parts with the latter obtained as

$$v_b = k_P e + k_I \int e dt, \tag{34}$$

in terms of a proportional (P) gain k_P and an integral (I) gain k_I. A consolidated voltage V_f is directly estimated in the CFFPI scheme, as shown in Fig. 2b. Besides a motive part V_{f0} that imparts oil flow between the paired valve and cylinder as the dominating part, Fig. 2c describes the OSFFPI composition as

$$V_f = V_{f0} + v_{fl} + v_{f\beta} \,:\, v_{fl} \ll V_f,\, v_{f\beta} \ll V_f, \tag{35}$$

where the relatively smaller compensation voltages are v_{fl} for the valve leakage and $v_{f\beta}$ for the compressibility-linked accumulation, referred hereafter also as compressible discharge.

In both the models, the pump discharge is modelled as

$$Q_p = Q_{a\dot{y}} + Q_{a\beta} + Q_l \,:\, Q_{a\beta} \ll Q_{a\dot{y}},\, Q_l \ll Q_{a\dot{y}}, \tag{36}$$

where a motive discharge imparting the piston motion is given by

$$Q_{a\dot{y}} = A_1 \max(\dot{y}_d, 0) - A_2 \min(\dot{y}_d, 0), \tag{37}$$

the compressible discharge in the cylinder chambers due to oil compressibility is expressed as

$$Q_{a\beta} = \{\, max(\dot{y}_d, 0)\dot{p}_{1f}\left(V_{01} + A_1 y_d\right) - min(\dot{y}_d, 0)\dot{p}_{2f}(V_{02} - A_2 y_d)\}/\beta. \tag{38}$$

and the leakage discharge Q_l is given by (6) and (7) for the CFFPI model and (8) and (9) in case of OSFFPI model. The subscript f implies feedforward estimation of pressures associated with the displacement demand and its derivatives. Corresponding to the notations used here, the pump pressure model (1) is rewritten as

$$p_{pf} = p_{ci}\left[2 + \coth\{(Q_{a\dot{y}} + Q_{a\beta} + Q_l - Q_{p0})/Q_c\}\right]. \tag{39}$$

In terms of this general description, the two controller designs are separately explained next.

4 Conventional feedforward controller design

The conventional feedforward, or CFF, controller shown in Fig. 2b is designed in terms of three modules.

4.1 Discharge estimator

Using (36), the summative estimation of the discharge is executed as follows.

4.1.1 Motive discharge

For the solenoid connections mentioned prior to (16), the discharges in the two cylinder chambers are estimated from (12–15) along with their relation with the dominant FF voltage as

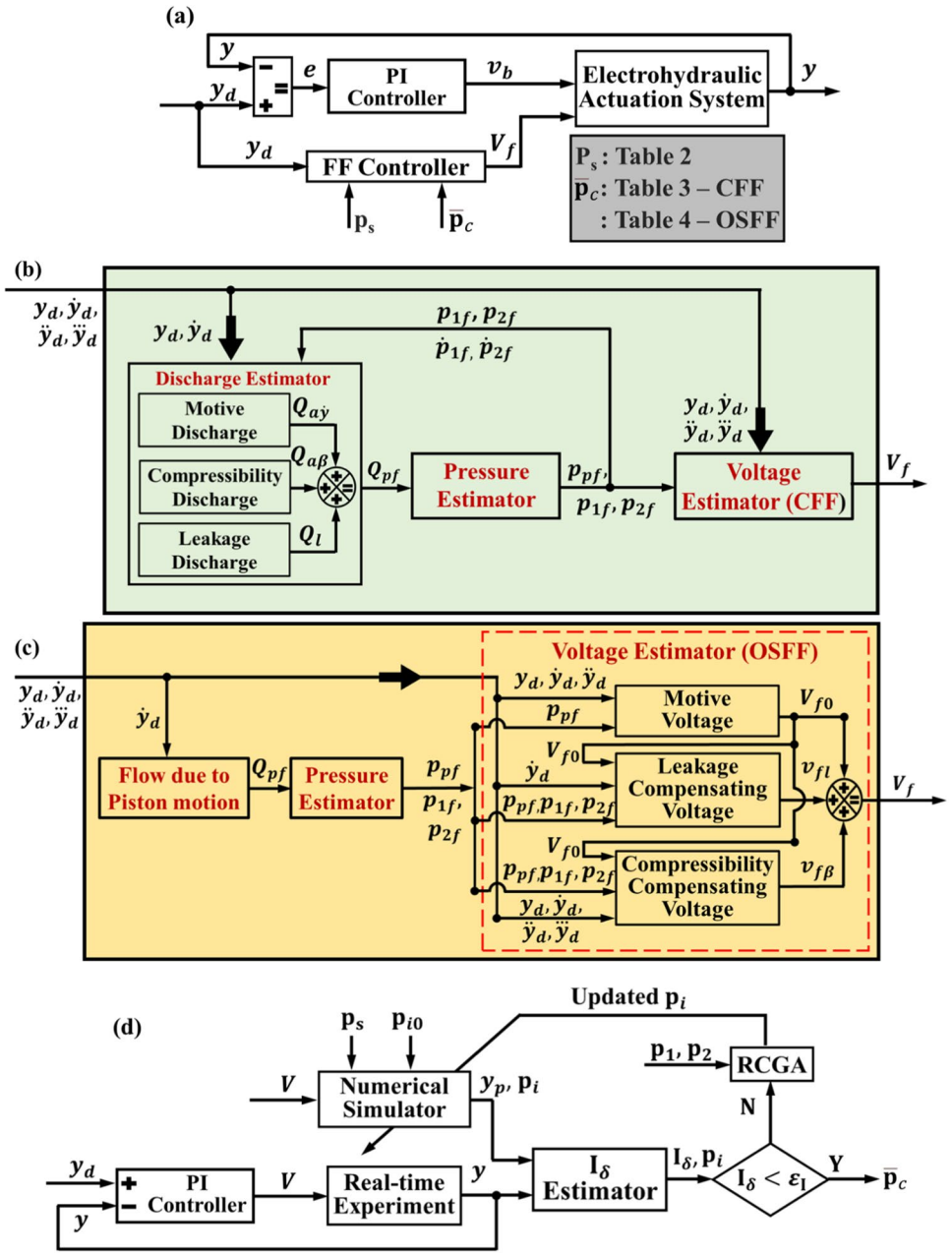


Fig. 2 Block diagrams for **a** FFPI-based motion control of electrohydraulic actuation system; **b** CFF voltage estimation; **c** OSFF voltage estimation and **d** parameter identification

$$A_1\dot{y}_d = Q_{1f} = c_{v1}V_f\sqrt{p_{pf}-p_{1f}} \text{ and}$$
$$A_2\dot{y}_d = Q_{2f} = c_{v2}V_f\sqrt{p_{2f}} \text{ for } \dot{y}_d \geq 0, \tag{40}$$

or

$$A_1\dot{y}_d = Q_{1f} = c_{v1}V_f\sqrt{p_{1f}} \text{ and}$$
$$A_2\dot{y}_d = Q_{2f} = c_{v2}V_f\sqrt{p_{pf}-p_{2f}} \text{ for } \dot{y}_d < 0. \tag{41}$$

4.1.2 Compressible and leakage discharges

Both the compressibility and leakage discharges require updated feedback of the pressure estimator. While the estimation of the compressibility discharge from (38) requires the derivatives of the pressures of the two cylinder chambers, the leakage discharge requires the updated pressures in (6) and (7).

4.2 Pressure estimator

A key assumption corresponding to (40) and (41) is the closely matched ratios of the port cuts in the valve sleeve and the flow areas in the connected cylinder chambers given by

$$c_{v2}/c_{v1} = A_2/A_1, \tag{42}$$

which leads to the condition of nearly equal pressure drops in the two metered ports leading to (19). The assumed pairing of feedforward pressures with the demand in (2) and (4) with the pertinent pair among (26–29) together with (40) and (41) lead to a sequential estimation of the pressures in the two cylinder chambers and the solenoid voltages. The chamber pressures are estimated as

$$\begin{aligned} p_{1f} =&[m\ddot{y}_d + ky_d + \{F_c + (F_s - F_c)\exp(-\dot{y}_d^2/v_s^2)\} \\ &\operatorname{sgn}(\dot{y}_d) + \alpha_v\dot{y}_d + p_{pf}A_2]/(A_1 + A_2), \end{aligned} \tag{43}$$

and

$$\begin{aligned} p_{2f} =&[p_{pf}A_1 - m\ddot{y}_d - ky_d - \{F_c + (F_s - F_c) \\ &\exp(-\dot{y}_d^2/v_s^2)\}\operatorname{sgn}(\dot{y}_d) - \alpha_v\dot{y}_d]/(A_1 + A_2). \end{aligned} \tag{44}$$

By combining the estimates of the above two in two consecutive time steps, the process of estimating the pressure derivatives as

$$\dot{p}_{1f} = (p_{1f}|_t - p_{1f}|_{t-\delta t})/\delta t, \tag{45}$$

and

$$\dot{p}_{2f} = (p_{pf} - p_{1f} - p_{2f})/\delta t, \tag{46}$$

calls for supplementation of (43) and (44) at the initial time. For this purpose, the pump pressure is obtained as p_{pf0} from (39) by approximating (36) only to be due to (37). Then by combining the no-load stationary piston condition in (2) with (19), the initial chamber pressures are evaluated as

$$p_{1f0} = p_{pf0}A_2/(A_1 + A_2) \text{ and } p_{2f0} = p_{pf0}A_1/(A_1 + A_2). \tag{47}$$

4.3 Voltage estimator

The appropriate combination of (40), (41), (42) and (43) gives the solenoid voltage estimation as

$$\begin{aligned} V_f = &\left[m\ddot{y}_d - \left\{2\dot{y}_d\ddot{y}_d(F_s - F_c)/v_s^2\right\}\exp(-\dot{y}_d^2/v_s^2)\operatorname{sgn}(\dot{y}_d) + \alpha_v\ddot{y}_d + k\dot{y}_d + \beta\dot{y}_d\{A_1^2/(V_{01} + A_1y_d)\right. \\ &+A_2^2/(V_{02} - A_2y_d)\} + \beta A_1\{Q_1^l/(V_{01} + A_1y_d) + \lambda Q_2^l/(V_{02} - A_2y_d)\}]/[\beta A_1c_{v1}\{1/(V_{01} \\ &\left.+A_1y_d) + \lambda^2/(V_{02} - A_2y_d)\}\{\max < \operatorname{sgn}(\dot{y}_d), 0 > (p_{pf} - p_{1f}) + \min < \operatorname{sgn}(\dot{y}_d), 0 > p_{1f}\}^{1/2}\right]. \end{aligned} \tag{48}$$

The formulation of the OSFFPI design is given in the next section.

5 Order-separated feedforward controller design

OSFFPI scheme employs the friction model (5) in place of (4) in the CFFPI scheme and the leakage model (8) and (9) instead of (6) and (7). Also, a consolidated feedforward voltage formulation is substituted by a split form (35) that dispenses the matching constraint (42). In comparison with the major motive voltage V_{f0}, relatively smaller voltages $v_{f\ell}$ compensate the leakage and $v_{f\beta}$ the discharges $q_{1\beta}$ and $q_{2\beta}$ at the cap and rod-side chambers of the cylinder, respectively, so as to write (12–15) as

$$\begin{aligned} Q_{1f} &= A_1\dot{y}_d + q_{1\beta} = c_{v1}\left(V_{f0} + v_{f\ell} + v_{f\beta}\right) \\ &\sqrt{p_{pf} - p_{1f}} - c_{l1}\sqrt{p_{1f}} \text{ for } \dot{y}_d \geq 0, \end{aligned} \tag{49}$$

$$\begin{aligned} Q_{2f} =&\lambda A_1\dot{y}_d + q_{2\beta} = c_{v2}\left(V_{f0} + v_{f\ell} + v_{f\beta}\right) \\ &\sqrt{p_{2f}} - c_{l2}\sqrt{p_{pf} - p_{2f}} \text{ for } \dot{y}_d \geq 0, \end{aligned} \tag{50}$$

$$Q_{1f} = A_1\dot{y}_d + q_{1\beta} = c_{v1}\left(V_{f0} + v_{f\ell} + v_{f\beta}\right)\sqrt{p_{1f}} - c_{l1}\sqrt{p_{pf}} \text{ for } \dot{y}_d < 0, \tag{51}$$

and

$$\begin{aligned} Q_{2f} =&\lambda A_1\dot{y}_d + q_{2\beta} = c_{v2}\left(V_{f0} + v_{f\ell} + v_{f\beta}\right) \\ &\sqrt{p_{pf} - p_{2f}} - c_{l2}\sqrt{p_{2f}} \text{ for } \dot{y}_d < 0. \end{aligned} \tag{52}$$

On the basis of this model, the estimator blocks depicted in Fig. 2c are constructed.

5.1 Pressure estimator

By invoking negligible oil leakage and compressibility contribution in the dominant motion inducing feedforward part in (2), (39) and (49–52), the expressions for the chamber pressures in the cylinder and the delivery pressure of the pump are estimated as:

$$p_{1f}|_{\dot{y}_d\geq 0} = p_{pf} - \{\dot{y}_d A_1/c_{v1}V_{f0}\}^2,\ p_{2f}|_{\dot{y}_d\geq 0} = \{\dot{y}_d \lambda A_1/c_{v2}V_{f0}\}^2, \tag{53}$$

$$p_{1f}|_{\dot{y}_d<0} = \{\dot{y}_d A_1/c_{v1}V_{f0}\}^2,\ p_{2f}|_{\dot{y}_d<0} = p_{pf} - \{\dot{y}_d \lambda A_1/c_{v2}V_{f0}\}^2, \tag{54}$$

$$p_{pf}|_{\dot{y}_d\geq 0} = p_{ci}[2 + \coth\{(A_1\dot{y}_d - Q_{p0})/Q_c\}], \tag{55}$$

and

$$p_{pf}|_{\dot{y}_d<0} = p_{ci}[2 + \coth\{(-\lambda A_1\dot{y}_d - Q_{p0})/Q_c\}]. \tag{56}$$

5.2 Voltage estimator

5.2.1 Motive voltage

In continuation of the above formulation, the motive voltage is estimated as

$$V_{f0}|_{\dot{y}_d\geq 0} = +A_1^3\left[\left\{\dot{y}_d^2\left(1/\left(c_{v1}^2\right) + \lambda^3/c_{v2}^2\right)\right\}/\left(p_{pf}A_1 - m\ddot{y}_d - ky_d - F_c - \alpha_v\dot{y}_d\right)\right]^{1/2}, \tag{57}$$

and

$$[V_{f0}|_{\dot{y}_d<0} = -A_1^3\left[\left\{\dot{y}_d^2\left(1/\left(c_{v1}^2\right) + \lambda^3/c_{v2}^2\right)\right\}/\left(m\ddot{y}_d + ky_d - F_c + \alpha_v\dot{y}_d - p_{pf}\lambda A_1\right)\right]^{1/2}. \tag{58}$$

5.2.2 Compressibility compensating voltage

The model for the compressibility-compensating voltage is formulated as

$$\begin{aligned} v_{f\beta} =& \operatorname{sgn}\left(\dot{y}_d\right)\left(m\ddot{y}_d + k\dot{y}_d + \alpha_v\ddot{y}_d\right) \\ & /\Big[\beta\Big\{c_{v1}A_1\{\max(\dot{y}_d,0)\sqrt{p_{pf}-p_{1f}} \\ & -A_2\min(\dot{y}_d,0)\sqrt{p_{1f}}\Big\}/\left(V_{01} + A_1y_d\right) + \beta c_{v2}\lambda A_1 \\ & \left\{\max(\dot{y}_d,0)\sqrt{p_{2f}} - A_2\min(\dot{y}_d,0)\sqrt{p_{pf}-p_{2f}}\right\}/ \\ & \left(V_{02} - \lambda A_1y_d\right)\Big], \end{aligned} \tag{59}$$

by using (35), (49) and (50) in (30) for the extension demand and (35), (51) and (52) in (31) for the retraction demand.

5.2.3 Leakage compensating voltage

By grouping the leakage-dependent terms in (49–52), the expressions for the leakage-compensating voltages are obtained as

$$\begin{aligned} v_{fl}|_{\dot{y}_d\geq 0} =& \{(c_{l1}/c_{v1})\sqrt{p_{1f}}/\sqrt{p_{pf}-p_{1f}} \\ & + (c_{l2}/c_{v2})\sqrt{p_{pf}-p_{2f}}/\sqrt{p_{2f}}\}/2, \end{aligned} \tag{60}$$

and

$$\begin{aligned} v_{fl}|_{\dot{y}_d<0} =& \{(c_{l1}/c_{v1})\sqrt{p_{pf}-p_{1f}}/\sqrt{p_{1f}} \\ & + (c_{l2}/c_{v2})\sqrt{p_{2f}}/\sqrt{p_{pf}-p_{2f}}\}/2. \end{aligned} \tag{61}$$

It is clear from (42) to (47) and (53–61) that the implementation of CFFPI and OSFFPI controllers in real time calls for identifying certain parameters, the procedure for which is detailed in the next section.

6 Parameter Identification Procedure

A real-coded genetic algorithm [34], or RCGA, targets minimization of a deviation index

$$\mathrm{I}_\delta = \int_0^{nT} |y_p(t) - y(t)|dt, \tag{62}$$

where Fig. 2d shows y as the piston displacement response in a real-time experiment with a PI controller and y_p stands for the open-loop prediction of the numerical simulator corresponding to the same voltage variation V, noted in the experiment and a randomly initialized parameter vector $\mathbf{p_i}$ that is constructed as

$$\mathbf{p}|_{\mathrm{CFFPI}} = (F_s, F_c, v_s, \alpha_v, V_{01}, V_{02}, Q_{l0}, Q_{l1}, \alpha_l, p_{ci}, Q_c)^T, \tag{63}$$

$$\mathbf{p}|_{\mathrm{OSFFPI}} = (F_c, \alpha_v, V_{01}, V_{02}, Q_c, p_{\mathrm{ci}})^T. \tag{64}$$

For the system parameter $\mathbf{p}_s$ given in Table 3 [37, 38], (62) has been used by setting n equal to 3 and the time period T

Table 3 System parameters $\mathrm{p}_s = (A_1, \lambda, \beta, m, k, c_{v1}, c_{l1}, \Omega, Q_{p0})$

Symbol	A_1	λ	β	m	k	$c_{v1} = 2\,c_{v2}$	$c_{l1} = 2c_{l2}$	Ω	Q_{p0}
Unit	$10^{-3}\mathrm{m}^2$	–	10^9 MPa	Kg	N/mm	$10^{-8}\mathrm{m}^{3.5}\,\mathrm{V}^{-1}\,\mathrm{kg}^{-1/2}$	$10^{-9}\mathrm{m}^{3.5}\,\mathrm{kg}^{-1/2}$	–	$10^{-4}\mathrm{m}^3\mathrm{s}^{-1}$
Value	1.256	0.75	1.3	2.5	20	3.46	3.8	0.66	8.3

equal to 0.67s to evaluate the integral for amplitude a of 20 mm in a sinusoidal displacement demand

$$y_d = a \sin(2\pi t/T + 3\pi/2) + a. \tag{65}$$

The sole experiment involves only a PI controller with a P gain of 180 V/m and I gain of 100 V/(m-s). In each predictive simulation, the pertinent feedforward model has been used in open loop with the variation of the input voltage with time set identical to that logged during the experiment. In the present work, the entire minimization process

$$\mathbf{p}_{ci} = \min_{\mathbf{p} \in (\mathbf{p_1}, \mathbf{p_2})} \mathrm{I}_\delta \text{ for } i = 1 \text{ to } 50, \tag{66}$$

involving (62) has been executed in the offline mode for fifty different randomly initialized $\mathbf{p}_{i0}$ between the specified lower bound $\mathbf{p_1}$ and upper bound $\mathbf{p_2}$ beginning with the first generation selected randomly between the vector bounds. The mean of the converged parameter vector

$$\overline{\mathbf{p}}_c = \left(\sum_{i=1}^{50} \mathbf{p}_{ci}\right)/50, \tag{67}$$

is considered as the final estimation involving a standard deviation

$$\mathbf{SD} = 100\left[\left\{\sum_{i=1}^{50} (\mathbf{p}_{ci} - \overline{\mathbf{p}}_c)(\mathbf{p}_{ci} - \overline{\mathbf{p}}_c)^T\right\}/50\right]^{1/2}/\overline{\mathbf{p}}_c\Big]. \tag{68}$$

The subsequent step to the initialization is the selection of the fittest candidates that has been achieved here by Roulette wheel selection method. A parameter vector stands higher probability of selection for multiple times, if it returns the lowest I_δ through (62). In the next step of BLX-α crossover that has been randomly executed by setting α equal to 0.5, five pairs are randomly formed out of the ten candidate surviving the fitness test. The random execution is enabled, if a selected random value falls above the crossover-enabling set value of 0.6. For an enabled pair, the BLX-α crossover expands the parameter range on each side by half of the difference of the respective scalar values in the vector pair subjected to the constraint of the lower and upper bounds. The crossover is executed in terms of two selected random values, each returning an interpolated parameter value within the expanded range. A non-uniform mutation is attempted in the final step, with its probability of execution reducing as the solution progresses over generations to convergence as the change of minimum I_δ falls below an ε_1 limit. The simulation runs have been performed in **Matlab-Simulink** with ***ODE1 Euler solver*** with a fixed time step of 10^{-6} s with the corresponding results discussed next.

7 Estimation of feedforward parameters

Tables 4 and 5, which pertain respectively to the CFFPI and OSFFPI controllers, contain the bounds in (66), the converged parameter vector (67) and the standard deviation (68). In Fig. 3, the convergence patterns of the minimum I_δ value, defined by (62) for both the conventional and order-separated feedforward parameter sets given, respectively, by (63) and (64) are shown. The observed initial fluctuations of I_δ may be attributed to the non-uniform mutation employed here, whose probability of relocating a parameter away from the expected good locations after crossover keeps on reducing over later generations.

Table 4 Bounds and converged system parameters for conventional feedforward model

Symbol	F_c	F_s	α_v	v_s	V_{01}	V_{02}	Q_{l0}	Q_{l1}	α_l	Q_c	p_{ci}
Unit	N	N	N s m^{-1}	mm s^{-1}	10^{-4}m^3	10^{-4}m^3	10^{-5}m^3/s	10^{-5}m^3/s	–	10^{-5}m^3/s	MPa
p_1	45.0	70.0	400.0	1.0	3.00	2.80	2.50	0.415	0.0008	1.25	12.00
p_2	60.0	90.0	550.0	7.0	3.50	3.20	4.17	1.250	0.0020	3.45	13.00
$\overline{p}_c$	55.0	85.0	482.0	5.17	3.40	3.10	3.72	1.150	0.0015	2.50	12.77
SD	2.92	2.85	13.17	0.453	0.2437	0.1382	0.16	0.00296	0.00023	0.167	0.153
SD (%mean)	4.69	3.35	2.73	8.31	7.17	4.45	4.30	0.257	15.33	6.68	1.19

Table 5 Bounds and converged system parameters for order-separated model

Symbol Unit	F_c N	α_v N s m^{-1}	V_{01} 10^{-4}m^3	V_{02} 10^{-4}m^3	Q_c 10^{-4}m^{3}s^{-1}	p_{ci} MPa
Lower bound,p_1	45.0	400.0	3.00	2.80	1.25	12.00
Upper bound,p_2	60.0	550.0	3.50	3.20	3.45	13.00
Converged mean,$\overline{p}_c$	53.1	461.6	3.30	3.05	1.70	12.80
SD	2.75	14.77	0.115	0.163	0.172	0.132
SD (%mean)	5.14	3.2	3.68	5.25	8.6	1.03

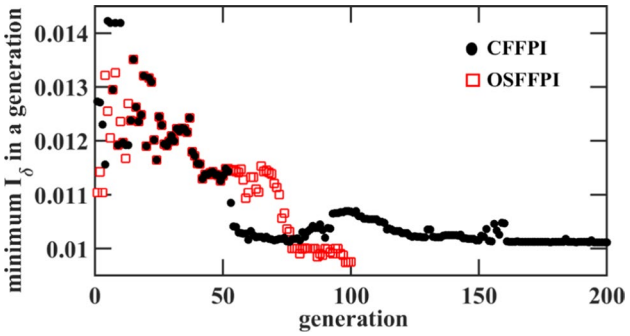


Fig. 3 Parameter estimation for the proposed feedforward models

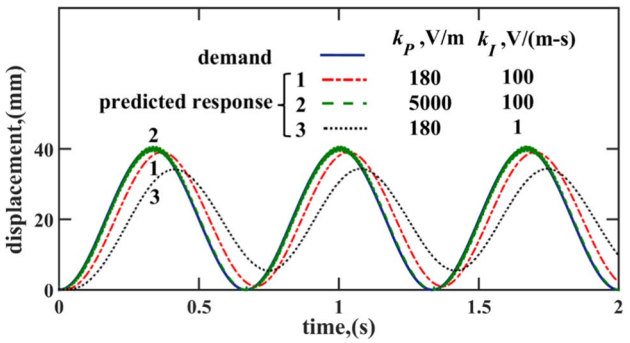


Fig. 4 Predicted variations of responses with time for different gains

Relatively higher enablement of mutation during the initial stage allows exploring newer part of the domain containing the global optimum. However, the occasional increase in I_δ in a later generation suggests the relocation to yield poorer solutions prior to coming to a matured stage. Much faster convergence evident in Fig. 3 is presumably due to lesser number of six OSFF parameters in (64) in comparison with eleven CFF parameters in (63).

Despite the differences in the parameter sets in (63) and (64) for the two schemes, all the converged values of all the six parameters except Q_c are close to those in Table 4, sometimes even falling within the limit of data truncation. Even these two substantially different Q_c give rise to two characteristic curves corresponding to (1) with only very localized difference in Q in the vicinity of the sharpest hyperbolic variation near p_{ci}. These observations indeed establish the identification scheme used here to be quite robust.

In a simulation run, the mean of the lower and upper bounds given in Table 4 for each parameter has been used to predict the time variation y_p for the piston displacement. These plots are presented in Fig. 4 for three widely differing gain combinations. Clearly, the predictions remain bounded well within safe limit of 200 mm strokes of both the cylinder and the **LVDT** of the set-up shown in Fig. 1. A safe containment of the stroking demonstrated in the figure provides confidence for executing an experimental identification run in real time by using those gain values.

Despite a better mean-parameter prediction revealed in Fig. 4 for a higher proportional gain of 5000 V/m, the lower proportional gain of 180 V/m in combination with 100 V/(m-s) integral gain has been used as the default setting so as to avoid any voltage saturation during the stability analysis and the detailed experimental study. The choice is strongly governed by its much closer prediction to the best combination of 5000 V/m and 100 V/(m-s) than that predicted for the combination of 180 V/m and 1 V/(m-s). In view of the third-order motion dynamics apparent in (30) and (31) and the PI controller, the overall system attains a fourth-order structure. Any order higher than two calls for a stability analysis over and above the positivity of the coefficients. The next section details the proposed stability approach.

8 Proposed approach for controller stability assessment

The backbone of the stability analysis is formed by the functions f_i with i equal to 1–5, where (1) introduces f_1, the pair (16) and (17) f_2, the pair (22) and (23) f_3, the pair (24) and (25) f_4 and the pair (30) and (31) f_5. Their generalized derivative is expressed as

$$f_{i_x} = \partial f_i/\partial x\big|_{(y_d,\dot{y}_d,\ddot{y}_d,p_{pf},p_{1f},p_{2f},V_f)} \text{ with } x = y,\ \dot{y},\ \ddot{y},\ p_p,\ p_1,\ p_2 \text{ and } V, \tag{69}$$

for composing a linearized model about any operating point $(y_d, \dot{y}_d, \ddot{y}_d, p_{pf}, p_{1f}, p_{2f}, V_f)$ by neglecting the terms beyond the first order in Taylor series expansion of (69). Of course, the first-order derivatives are related to the perturbation about the demand and about the feedforward component for the other variables. From (30) and (31), the equation for the error dynamics is thus expressed as

$$\begin{aligned}\ddot{e} \approx \dddot{y}_d - \ddot{y} = f_{5_y}\ddot{e} + f_{5_{\dot{y}}}\dot{e} \\ + f_{5_y}e + f_{5_V}v_b + f_{5_{p_p}}p_p + f_{5_{p_1}}p_1 + f_{5_{p_2}}p_2.\end{aligned} \tag{70}$$

Similar treatment for the pair (16) and (17) and then for (1) yields the perturbation of the discharge and pressure at the pump delivery as

$$q_p \equiv Q_p - Q_{pf} \approx f_{2_V}v_b + f_{2_{p_p}}p_p + f_{2_{p_1}}p_1 + f_{2_{p_2}}p_2, \tag{71}$$

and

$$p_p = p_p - p_{pf} \approx f_{1_{Q_p}}q_p. \tag{72}$$

In the same manner, the pressure perturbations in the cylinder are obtained from the pair (22) and (23) for the cap-end

chamber and from the pair (24) and (25) for the rod-end chamber, respectively, as

$$p_1 = p_1 - p_{1f} \approx f_{3_{p_p}} p_p - f_{3_{\ddot{y}}} \ddot{e} - f_{3_{\dot{y}}} \dot{e} - f_{3_y} e, \tag{73}$$

and

$$p_2 = p_2 - p_{2f} \approx f_{4_{p_p}} p_p - f_{4_{\ddot{y}}} \ddot{e} - f_{4_{\dot{y}}} \dot{e} - f_{4_y} e. \tag{74}$$

Eliminating v_b, p_p, p_1 and p_2 from (70) by using (32) and (72–74), the error dynamics is written as

$$f(t) \equiv \dddot{e} + a_2 \ddot{e} + a_1 \dot{e} + a_0 e + a_{-1} \int e dt = \delta, \tag{75}$$

where the expressions for the coefficients a_2, a_1, a_0 and a_{-1}, respectively, with dimensions s^{-1}, s^{-2}, s^{-3} and s^{-4} are given in Appendix and δ stands for the neglected higher-order terms. Laplace transform of the above equation yields the characteristic polynomial and the characteristic equation respectively as

$$\int f(t) \exp(-st) dt \equiv F(s)[\int \{(\int e dt) \exp(-st)\} dt], \tag{76}$$

and

$$F(s) \equiv s^4 + a_2 s^3 + a_1 s^2 + a_0 s + a_{-1} = 0 \text{ with } (a_2, a_1, a_0, a_{-1}) > 0, \tag{77}$$

$$\text{where} s = \sigma + j\omega \text{ with } j = \sqrt{-1}, \tag{78}$$

along with the conditions in (77) need to be fulfilled by positive values for the coefficients at every operating point. The characteristic Eq. (77) is of fourth-order, Routh array for which is expressed as

$$R = [r_{ij}] \text{ with } i = 1 \text{ to } n \text{ and } j = 1 \text{ to } (n+2)/2, \tag{79}$$

with n equal to 4 in (79). The coefficients in the first two rows of (79) are used to form an even and odd pair of polynomials respectively as

$$F_4(s) = s^4 + a_1 s^2 + a_{-1} s^0, \tag{80}$$

and

$$F_3(s) = a_2 s^3 + a_0 s. \tag{81}$$

Next, the coefficients of the subsequent alternate even and odd polynomials of reducing orders of s are written in the manner, for instance to that of Meinsma [41], as

$$r_{kj}(s) = r_{k-2,2} - \left(r_{k-2,1}/r_{k-1,1}\right) r_{k-1,2} \text{ for } 3 \le k \le (n+1) \text{ and } 1 \le j \le (n+2)/2, \tag{82}$$

along with the corresponding polynomials for each k in one of the conditions above

$$F_{n-k+1}(s) = \sum_{j=1}^{(n+2)/2} \left[\left\{r_{k-2,2} - \left(r_{k-2,1}/r_{k-1,1}\right) r_{k-1,2}\right\} s^{n-k+3-2j}\right], \tag{83}$$

yielding the first column of Routh table [41] as

$$C_1 \equiv \begin{pmatrix} 1 & a_2 & r_{31} = a_1 - a_0/a_2 \\ & r_{41} = a_0 - a_2 a_{-1}/r_{31} & a_{-1} \end{pmatrix}. \tag{84}$$

The polynomial form (77) could be achieved by recombining (80) and (81) as one particular outcome $Q_{1,0}(s)$ with k equal to 1 and μ equal to 0 in the family described in terms of the integer k and the real-valued μ defining

$$Q_{k,}(s) = F_{n-k+1}(s) + (1 - s) F_{n-k}(s) \text{ with } 1 \le k \le n \text{ and } \in R. \tag{85}$$

In the family (85), $Q_{k,0}(s)$ represents a branch with μ equal to zero signifying the addition of two successive polynomials yielding

$$Q_{k,0}(s) = F_{n-k+1}(s) + F_{n-k}(s), \tag{86}$$

in which first summation $Q_{1,0}(s)$ of successive odd–even polynomials leads to (77). Of course, any characteristics polynomial with positive coefficients can be viewed to be composed of factors that contain the highest power of s either equal to 1 or 2 with positive real coefficients, and the solution of the quadratic factor equal to zero yields complex conjugate roots. The factor with the highest power s equal to one and positive coefficients indeed gives rise to a negative zero of the polynomial (77) implying its stability. Hence, the quadratic factors with positive coefficients and complex conjugate roots need attention for the assessment of the stability behaviour.

Clearly, the order of polynomial $Q_{k,\,\mu c1}(s)$ for

$$\mu_{ck} = r_{k1}/r_{k+1,1}, \tag{87}$$

is one less than to that of $Q_{k,0}(s)$ that is induced in view of (83) by the subtractive term in (85) for the value (87) along with the additive terms in (85) forming the polynomial (86). In other words, the orders of $Q_{k+1,\,\mu_{c1}}(s)$ and $Q_{k,0}(s)$ are identical. Hence, the entire process of reduction of k from n in steps of 1–0 through the branch $Q_{k,\mu}(s)$ with continuous change of μ from 0 to μ_{ck} given by (87) for each k is indeed continuous. The approach to $Q_{k+1,\,\mu c1}(s)$ can be viewed as the coefficient of the leading quadratic factor in $Q_{k+1,\,\mu}(s)$ approaching zero through a small ϵ. By rewriting the quadratic factor in the limit of a small ϵ and processing it shown next as

$$\epsilon s^2 + bs + c = 0 \approx \epsilon s^2 + (b + \epsilon c/b)s + c = 0 \Rightarrow s = -c/b, -b/\epsilon, \tag{88}$$

the movement of the second root to infinity through the left-half of σ-ω plane can be appreciated, if the ratio in (88) is positive. This implies migration of one zero in a quadratic

factor with negative real parts through the left-half plane all along. The pair $F_{n-k+1}(s)$ and $F_{n-k}(s)$ in (85) always remains an even–odd combination of successive orders, and their common zeros are zeros of $Q_{k,\mu}(s)$ and $Q_{k,0}(s)$ too. Consequent to the order reduction of $Q_{k,\mu}(s)$ to $Q_{k+1,\mu}(s)$ at their connection point at $\mu_{ck} = r_{k1}/r_{k+1,1}$, a continuous variation of μ with occasional discrete increase of k from one to n at μ_{ck} eventually yields a polynomial of order one with negative zero. Of course, a negative zero in a first-order characteristic equation implies a stable system. The form of (88) indicates that all migration takes place through the left-half plane for a positive coefficient vector involving those in (84), which together with (77) lead to the additional constraint

$$r_{31} > 0 \text{ and } r_{41} > 0. \tag{89}$$

Such a migration was envisaged by Meinsma [41] as a process within a continuous family. The compliance with (77) and (89) implies it to represent a stable system with all the zeros containing a negative real part. This continuity argument is extended now for the linear time-varying form of (77) to involve shift of the operating point as well at close intervals. This shift could be viewed as a continuous change in the coefficient values in (77) from one constant set to another at a rate slower than that of μ. As long as (77) and (89) are not violated, (87) clearly remains stable. This is an important finding of the analysis presented here. The analysis is completed by presenting the results of the PI gain selection for the two controllers in the next section.

9 Stability analysis for CFFPI and OSFFPI controllers

The controller stability is ensured by positive coefficients in (77) and two of their combinations defined in (84) and subsequently used in writing (89). Of course, (77) involves the same coefficients as in (75), which in turn emerges through manipulations of (70–73) comprised of five functional derivatives expressed by (69). Hence, the coefficients in (77) and (89) depend on the shifting operating point along the track of the demand and its derivatives along with the P and I gain in (34) that form the feedback part of the overall voltage (33) appearing in (69). Over the frequency range between 0.1 and 3 Hz in the demand form (65), the variations of these coefficients over a cycle with 20 mm amplitude are depicted in Figs. 5, 6, 7 for the CFFPI and OSFFPI controllers. Of course, the different nature of variations for the two controllers are justifiable due to the use of different friction models (4) and (5) in the CFFPI and OSFFPI controllers, respectively, along with leakage models (6) and (7) for the CFFPI case and (8) and (9) for the OSFFPI case. In all the figures, the variation of the coefficients beyond the initial time appears greater than zero for the combination of P gain of 180 V/m and I gain of 100 V/(m-s) indicating the fulfilment of the stability criteria.

In Fig. 5a and b, pertaining to the CFFPI controller, the strong influence of the friction model (4) on a_2 and a_1 is evident corresponding to the sharp rise of the static friction over a phase about each velocity reversal in the demand (65) that are apparent in Fig. 4 at the beginning, middle and end of a cycle. With increase in the frequency, the variation over these phases clearly becomes sharper. The subsequent changes of the trend observed in Fig. 5a in case of the three highest frequencies are consistent with the decreasing Stribeck friction involving the exponential term in (4) followed by linearly increasing fully lubricated viscous friction with increase in the piston speed [18]. In Fig. 5b, the influence of Stribeck friction appears weaker and the viscous friction non-existent presumably under the greater influence of the shifting operating point.

In Fig. 5e, the variation of a_2 under the OSFFPI controller seems to be sensitive to the interaction of the forces. A look at Fig. 1 reveals that the extension stroke is aided by the net thrust on the piston and opposed by the compression spring, whereas the retraction stroke is aided by both of these forces. The net effect is manifested in Fig. 5e as frequency-independent variation during the extension phase and progressive increase of the amplitude yielding lower minimum value up to 2 Hz frequency during the retraction phase. A comparison of Fig. 5b and f reveals the superimposition of the additional effect in Fig. 4b arising from the exponential Stribeck term in (4) for the CFFPI controller with respect to the friction model (5) of the OSFFPI controller.

In Fig. 5c and d, the variations of a_0 and a_{-1} under the CFFPI controller appear to be frequency-independent. These trends could be due to the strong influence remaining mute for keeping P gain constant in Fig. 5c and I gain in Fig. 5d, whereas (96) and (100) in the Appendix reveal direct influence of P gain on a_0 and (97) shows a similar effect of I gain on a_{-1}. In Fig. 5g and h, the influence of the variation of the operating point seems to have surfaced for the same selected gains in case of the OSFFPI controller. This is acceptable in view of the difference in the two formulations.

It is evident from (96) and (97) that higher positive a_0 and a_{-1} are attainable by higher choices of P and I gain, respectively. By inspecting (75), it can be inferred that in the absence of I gain, the steady error would be equal to δ/a_0 that would be lower for higher a_0 achievable with higher P gain beyond a threshold. The incentive of employing I gain embedded within a_{-1} could be identified from (75) as achieving zero steady error despite the uncertainties embodied within δ by simultaneous fulfilment of

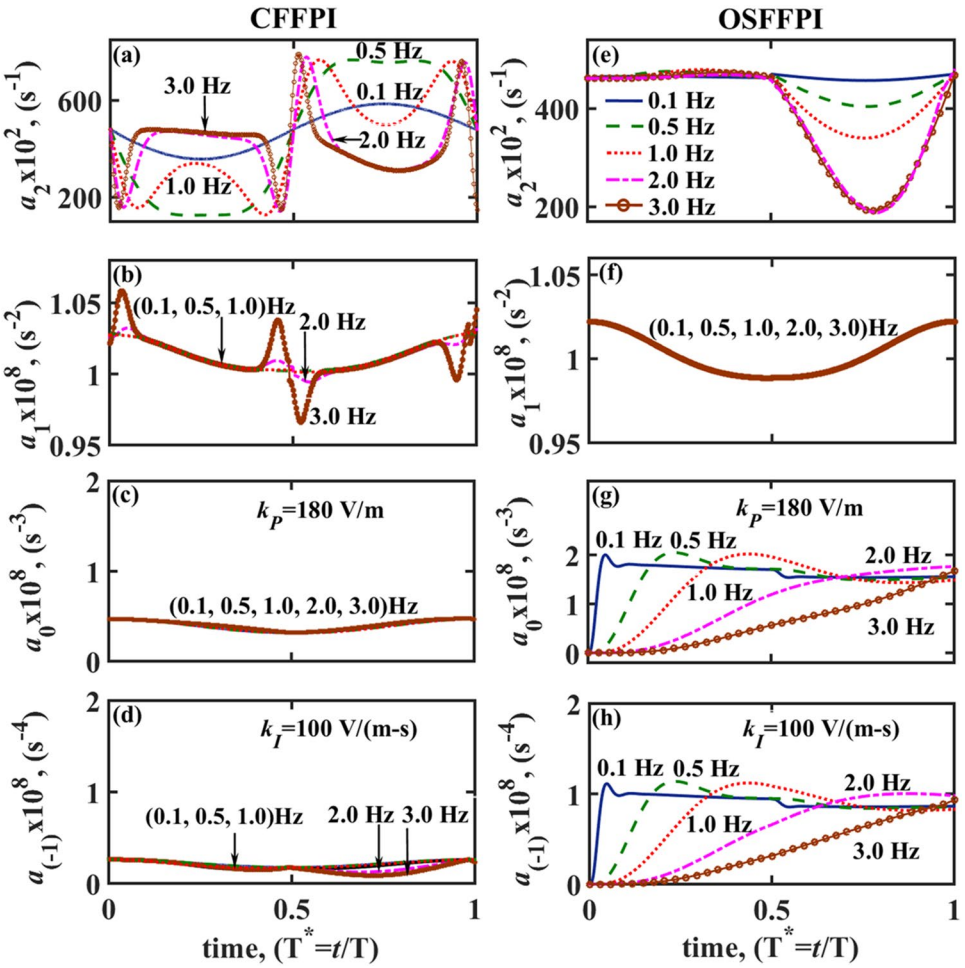


Fig. 5 Variation of coefficients of characteristic equations over sinusoidal demand cycles of different frequencies for CFFPI in **a–d** and OSFFPI in **e–h**

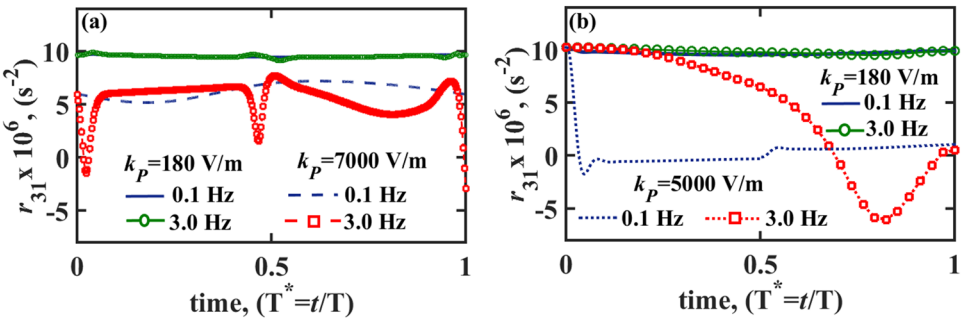


Fig. 6 Effect of proportional gain variation on r_{31} under **a** CFFPI and **b** OSFFPI controllers

$$a_{-1}\int_0^{t\geq t_s} e\,\mathrm{d}t = \delta \text{ and } e|_{t_s} = 0 \text{ for any } t \geq t_s, \tag{90}$$

where t_s stands for the settling time. Since the predicted performance in Fig. 4 reveals much improved performance for I gain of 100 V/(m-s) in comparison with 1 V/(m-s), the higher of the two I gains seems to be the choice for performing the real-time experiments. Hence, the effect of different P gains on the coefficients in (89) is studied next.

The variation of r_{31} over a cycle shown in Fig. 6 clearly shows the coefficient assuming negative values for high P gain for both the controllers. In Fig. 7, r_{41} remains positive throughout for both the gains. Hence for the sake of stability, the lower P gain of 100 V/m has been employed in all real-time experiments, the results of which are discussed in the next section.

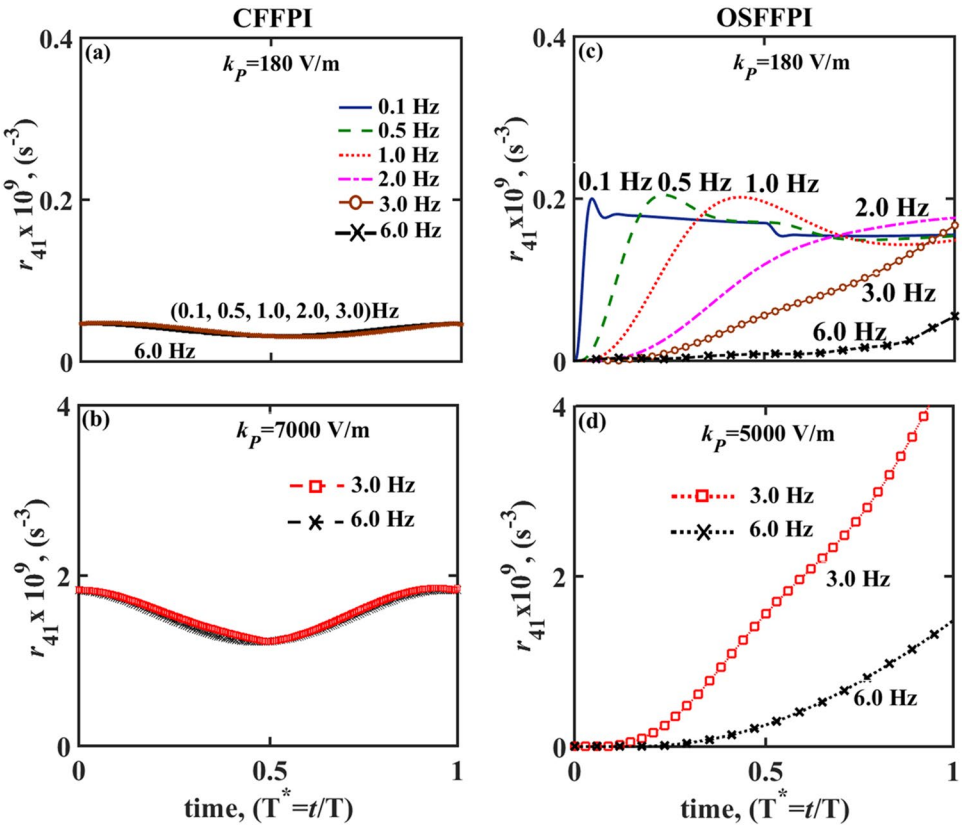


Fig. 7 Variation of r_{41} over a cycle for 100 V/(m-s) I gain and P gains of **a** 180 V/m and **b** 7000 V/m under CFFPI controller and **c** 180 V/m and **d** 5000 V/m under OSFFPI controller

10 Results and Discussion

Experimental results for sinusoidal position tracking demand (65) with 20 mm amplitude over frequency range from 0.1–9 Hz are presented in Figs. 8, 9, 10, 11, 12, 13, 14, 15 and Table 6 for the parameter values listed in Tables 3 and 5. These results provide a comparison of the performance of the OSFFPI controller with the CFFPI controller, an existing adaptive sliding mode controller involving a nonlinear integral controller, referred as ASMIC [30], and a simple PI-only controller. The pump has been operated in the fixed-displacement mode in case of the validation experiment against existing fixed-displacement mode results of ASMIC and in variable-displacement mode for the other three controllers. Table 6 lists the values of the integral absolute error

$$\mathrm{IAE} = \left(\int_0^{nT} |e|dt\right)/(anT), \tag{91}$$

and the integral square control effort

$$\mathrm{ISCE} = \left(\int_0^{nT} V^2 dt\right)/(V_{\max}^2 nT), \tag{92}$$

estimated over a period of three consecutive cycles by setting n equal to 3. Average power saved under both the OSFFPI and CFFPI controllers is shown in Fig. 15 that have been

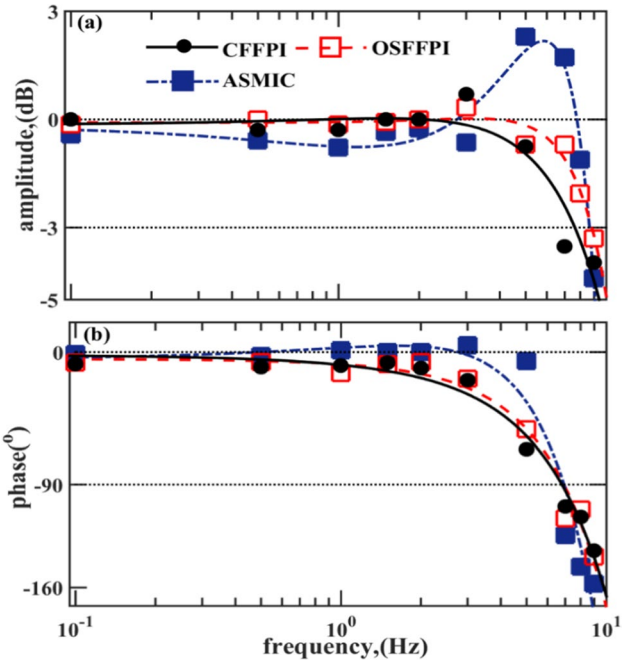


Fig. 8 Comparison of frequency responses of CFFPI, OSFFPI and an existing ASMIC controllers

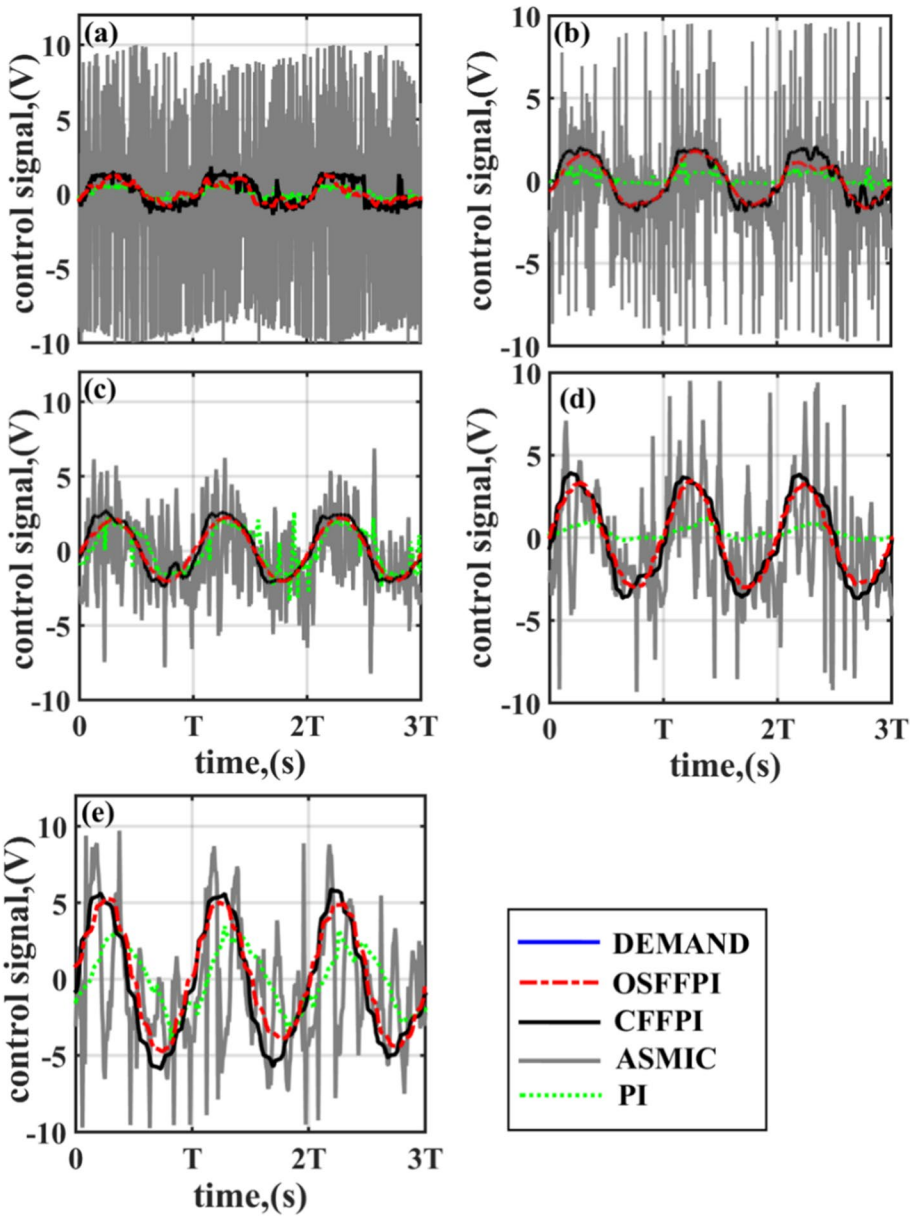


Fig. 9 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers for time variation of excitation voltage for a sinusoidal demand at frequency of **a** 0.1 Hz, **b** 0.5 Hz, **c** 1 Hz, **d** 2 Hz and **e** 3 Hz

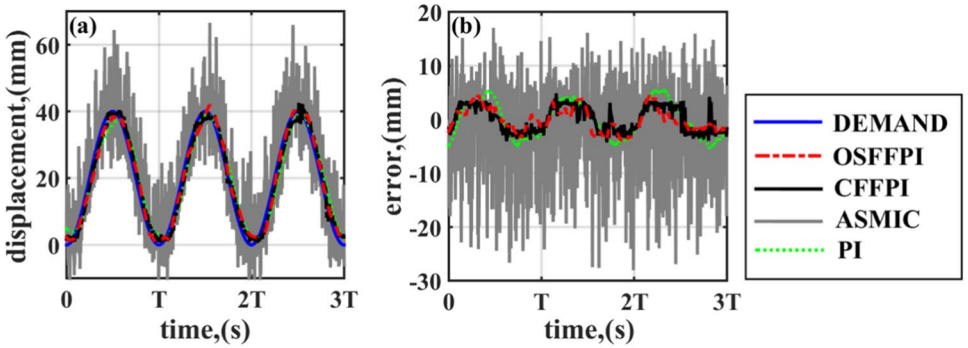


Fig. 10 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers at 0.1 Hz frequency of a sinusoidal demand for the **a** response and **b** error

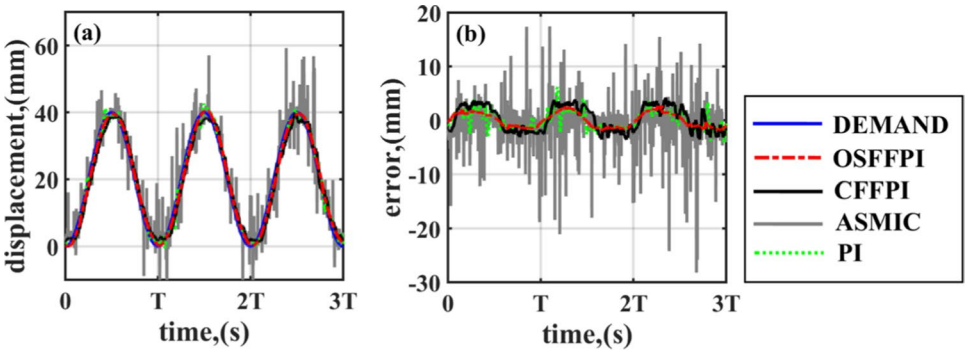


Fig. 11 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers at 0.5 Hz frequency of a sinusoidal demand for the **a** response and **b** error

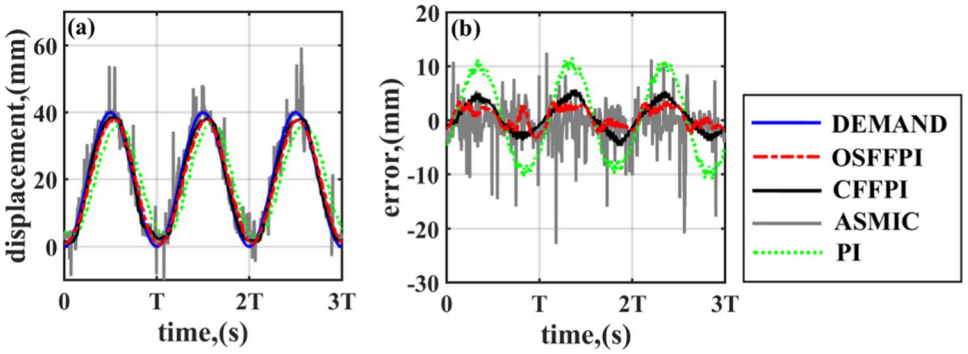


Fig. 12 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers at 1 Hz frequency of a sinusoidal demand for the **a** response and **b** error

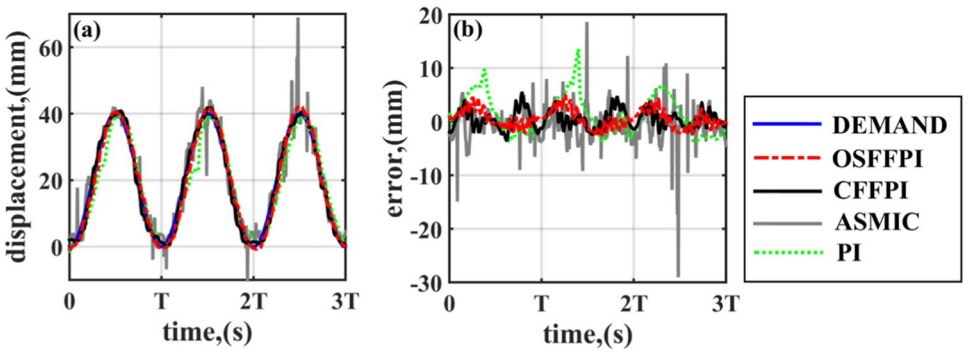


Fig. 13 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers at 2 Hz frequency of a sinusoidal demand for the **a** response and **b** error

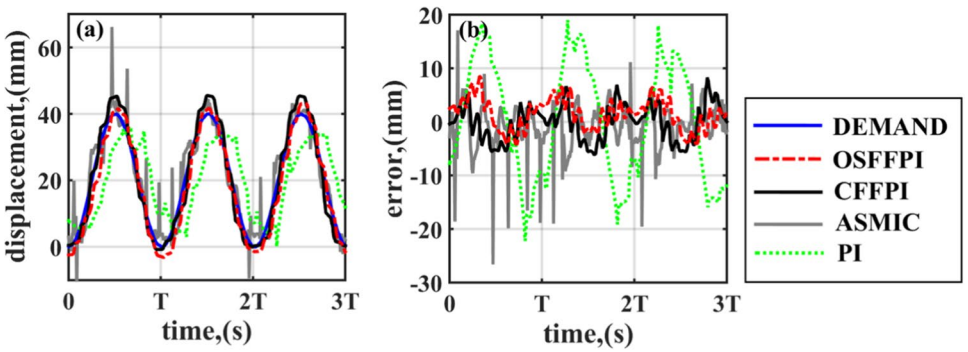


Fig. 14 Comparison of PI, CFFPI, OSFFPI and ASMIC controllers at 3 Hz frequency of a sinusoidal demand for the **a** response and **b** error

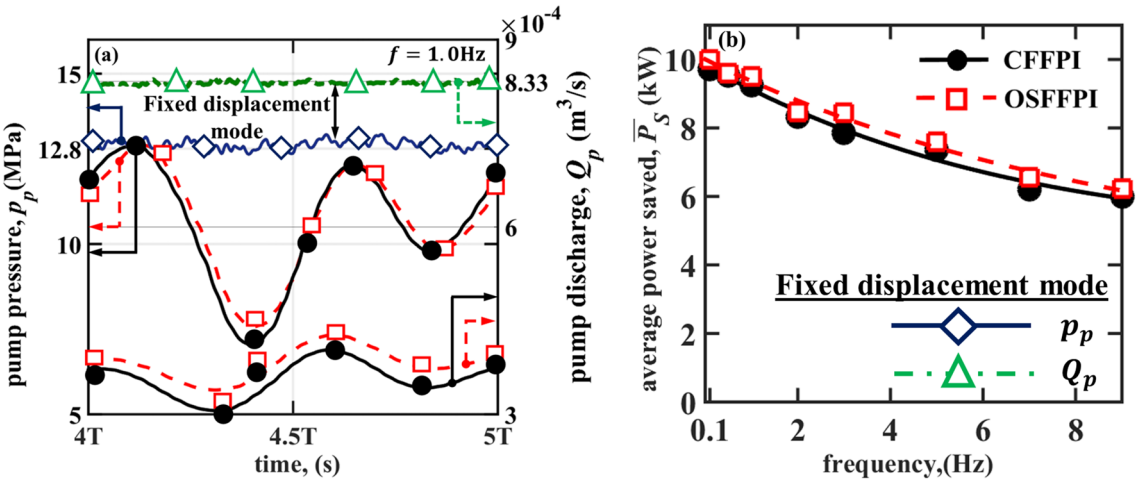


Fig. 15 Comparison of CFFPI and OSFFPI controllers on variable-displacement pump for **a** pressure and discharge variations over a cycle and **b** power saving over fixed displacement working

Table 6 Variation of performance indices with frequency for different controllers

Controller	CFFPI		OSFFPI		ASMIC		PI	
Demand frequency	IAE	ISCE	IAE	ISCE	IAE	ISCE	IAE	ISCE
0.1 Hz	**0.126**	**0.0074**	**0.124**	**0.0045**	0.124	0.0491	0.253	0.0064
0.5 Hz	**0.110**	**0.0184**	**0.086**	**0.0148**	0.133	0.0543	0.214	0.0014
1.0 Hz	**0.210**	**0.0310**	**0.182**	**0.0246**	0.142	0.0673	0.482	0.0187
2.0 Hz	**0.135**	**0.1045**	**0.107**	**0.0795**	0.197	0.1102	0.253	0.0025
3.0 Hz	**0.220**	**0.3821**	**0.212**	**0.1546**	0.232	0.1694	0.714	0.0353
4.0 Hz	**0.468**	**0.4123**	**0.267**	**0.2662**	0.270	0.3273	0.810	0.0543
5.0 Hz	**0.965**	**0.7517**	**0.884**	**0.6213**	0.532	0.5653	1.050	0.0733
7.0 Hz	**1.840**	**0.8450**	**1.835**	**1.2463**	2.157	0.9035	1.856	0.1672

achieved by operating the pump in the variable-displacement mode in comparison with the fixed-displacement mode maintaining the pump delivery at the relief valve pressure p_{RV}. The saving is calculated as

$$\overline{P_s} = Q_{p0}p_{RV} - (\int_0^T (Q_p p_p)dt/T). \tag{93}$$

In terms of these results, the contribution of the present work is explained next.

A demonstration of the acceptability of the OSFFPI controller is evident in Fig. 8 from the variations against the excitation frequency showing marginally lower deviation of the output amplitude from the demand and closely matching phase lag achieved under the CFFPI controller. Over the demand range between 0.4 Hz and 5 Hz, both the controllers exhibit lower amplitude deviation in comparison with that achieved under the existing ASMIC. Up to about 3 Hz demand, the lower phase shift achieved by both the proposed controllers is indeed an achievement offsetting the good effect of the marginally lower phase deviation under the ASMIC.

In Fig. 9, increased variation of the valve voltage with increase of the frequency demand reduces the pressure drops in its metering ports. Hence, the pressures gap in each of (12–15) for the two cylinder chambers with respect to the communicating pump or tank becomes less wide and the demanded acceleration variation of the piston is sustained following (2). The weakness of the existing ASMIC controller is evident in Figs. 9, 10, 11, 12, 13, 14 in its more oscillatory response than the other controllers. In the existing ASMIC, oscillations arise from time-to-time on–off switching of the SMC implemented in the vicinity of low error for chattering attenuation [30]. This oscillation is of much lower frequency in comparison with the chattering produced by a simple SMC. However, under the effect of this voltage oscillation of the ASMIC, Figs. 10, 11, 12, 13, 14 reveal corresponding oscillations in the response too. The oscillation-free excitation and response achieved under both the proposed FFPI scheme along with their lower IAE are indeed a notable contribution. The highest IAE in case of the

PI-only controller apparent in Table 6 with correspondingly the largest error variation with time exhibited in these figures justifies the use of more advanced controllers despite their higher implementation complexities.

The lowest IAE at 1 Hz noted in Table 6 in case of ASMIC seems to have arisen under the domination of the lower phase-lag effect, offsetting the effect of its marginally greater amplitude deviation apparent in Fig. 8. Near 2 Hz demand, all the three controllers in Fig. 8 exhibit insignificant amplitude difference between the output and the demand. Despite the least phase deviation of ASMIC in this case, its IAE in Table 6 appears higher than the IAE of each of the proposed two FFPI controllers. This higher IAE could be attributed to the significantly more large-amplitude oscillatory response of the ASMIC evident in Figs. 10, 11, 12, 13, 14. It is logical to accept the oscillations of much lower amplitude observed in Figs. 9, 10, 11, 12, 13, 14 for the two FFPI controllers as the usual overshoots and undershoots about the demand track arising under, I gain.

The lowest IAE under ASMIC, listed in Table 6 at 5 Hz with Fig. 8 exhibiting similar phase lag in all the cases, could be surmised as the domination of the effect of significantly lower amplitude deviation than the other two controllers offsetting the effect of more oscillatory ASMIC response. All the three controllers in Fig. 8 reveal bandwidth of about 6 and 7 Hz corresponding to 3 dB amplitude reduction and 90° phase lag, respectively. Table 6 clearly reveals a strength of the OSFFPI controller by its either nearly equal or lower IAE than those under CFFPI over the entire bandwidth range. Such an achievement at the expense of lower ISCE up to 5 Hz by the OSFFPI controller indeed establishes its superiority over the CFFPI controller.

In most cases of the frequency demand presented in Table 6, the expense of ISCE under the two FFPI controllers arising lower than the ASMIC is a testimony of the soundness of the feedforward models. In the Lyapunov function-based adaptation deployed in the ASMIC [30], only the minimization of the tracking error gets emphasized instead of the model-based estimation of the feedforward voltage in (33). Achieving the lowest IAE at the cost of the lowest ISCE by the ASMIC only at 5 Hz frequency is only fortuitous. In the existing SMC-based scheme [30], the adaptation of the SMC parameters is kept suspended near the origin of the phase plane comprised of the error (32) and its rate for mitigating the chattering effect. Such a scheme would have eliminated the chatter, if the trajectory would have remained contained within the stipulated bound forever after its first progression into the bound. But the uncertain disturbances would make the error to hover around zero and larger disturbances would have pushed the error outside the bound giving rise to a spike in the voltage excitation as well as in the response in each case. Such occurrences must have been low at 5 Hz leading to the corresponding best performance among all the controllers studied.

In Fig. 15, the pump operation in variable-displacement, or VD, mode highlights an important aspect of power saving achieved with respect to the pump operation in the fixed-displacement, or FD, mode. The performances of the two FFPI controllers proposed here have been compared corresponding to the demand form (65) for the piston displacement. In Fig. 15a, the variations of delivery pressure and discharge of the pump operating at both FD and VD modes over a full cycle for 1 Hz frequency demand are shown. For FD operation, the relief valve pressure p_{RV} has been set at 12.8 MPa very near to the identified cut-in value of the pump given in Table 5 along with lowering of the cut-in setting of the pre-compression spring of the pressure compensator valve of the pump so as to raise the cut-in value above the relief valve setting. The observed variations in Fig. 15a reveal nearly constant discharge Q_{p0} equal to $8.3\times10^{-4}\mathrm{m}^3/\mathrm{s}$ and pressure of 12.8 MPa along with low-amplitude ripples typical of FD operation without accumulator. These amplitudes are indeed insignificant with respect to those apparent during the executed two VD-mode operations.

For a VD mode operation, the cylinder indeed requires lower pump discharge to the rod-end chamber during a retraction stroking over a half cycle than during the extension stroking over a half cycle on either side to the larger-area cap-end chamber. Clearly, the VD-mode discharges in Fig. 15a indicate phase delays of about 45° with respect to the demand. This is apparent from the variations at the beginning of the fourth pumping cycle to match with ongoing later-phase piston retraction in the third cycle. During the beginning of the fourth cycle, Fig. 15a shows a falling pump discharge along with mostly a falling pump pressure. However, the discharge requirement to the cylinder during this phase goes on increasing. It seems that the lower compensation requirements to the motive discharge in the cylinder against oil compressibility and leakage due to the falling pressure enable the piston to sustain the motion demand quite well despite the falling pump discharge.

The inferred lower compensation requirements during the periods of lower variation of pump pressure indeed get corroborated by the reasonably good tracking at 1 Hz evident in Fig. 8b. For both the controller forms, the lags are insignificant in comparison with the phase difference between the pump discharge and the demanded requirement apparent in Fig. 15a. Higher tracking error for the CFFPI controller apparent in Fig. 12b with respect to the OSFFPI controller must have arisen due to the difference in the variation of the discharges in Fig. 15a for the two controller forms.

With respect to the extension stroking, the lower variation of the command excitation apparent in Fig. 9c during the retraction stroking implies smaller opening of the metered port pair of the valve and the consequent higher pressure

drop. This drop seems to have offset the effect of the compression spring force in (2) opposing the extension stroking and aiding the retraction stroking. A larger variation of pump pressure observed in Fig. 15a compensates despite providing lower retracting discharge for the rod-end cylinder of the cylinder. Of course, the small orifice-like openings of the metered ports give rise to large pressure drop. At the same time, a small variation of the excitation signal gives rise to a fast-response pressure variation that becomes essential for sustaining high-frequency motion demands. The pressure dynamics of the pump appears more sensitive than the discharge from the observed minimum pressure of about 7.5 MPa that is well below the cut-in limit of 12.8 MPa noted in Table 4 despite a continued discharge variation in apparent contradiction of the steady-state characteristics [37].

The observed average power saved shown in Fig. 15b for VD pumping arises from the elimination of the loss in the relief valve pressure in the FD mode. By ignoring the ripple effects as minor as apparent in Fig. 15a, the FD-mode power is calculated in the first part of (93) as equal to 10.6 kW corresponding to the nearly constant discharge Q_{p0} equal to $8.3\times10^{-4}\mathrm{m}^3/\mathrm{s}$ and the relief valve set at a pressure p_{RV} of 12.8 MPa. The differences from this power in the estimation by (93) appear comparable for both the proposed schemes. In fact, the relief valve setting for the pump operation in the FD mode has been set consistent with the identified cut-in value in Tables 4 and 5 so as to avoid penalization of the FD mode power by unduly high setting of the relief valve pressure. In VD-mode operations, the relief valve setting has been raised to a higher level so as to keep it closed during normal operation and protect the system only in case of any accidental over-pressurization. Nearly equal peak pressure in the VD mode and the average pressure in the FD mode evident in Fig. 15a is a testimony the achievement of such manual settings.

The return flow to the tank through the relief valve keeps on increasing with decrease in the frequency demand implying lower flow requirement for the motion actuation. Consequently, the average power saving in the VD mode reduces with increase in frequency demand. The observed power savings of about 9 kW at 1 Hz demand and 6 kW at 9 Hz demand imply the pumping power requirements in the VD mode to be, respectively, just 15 and 44% of 10.6 kW required in the FD mode. Higher power saving achieved by the more generalized OSFFPI scheme except for very low-frequency demand reveals a strength of this scheme.

11 Conclusions

The major contribution of this work is formulating a feedforward controller by exploiting the naturally emerging higher-order motive flow in comparison with leakage and compressible flows in an electrohydraulic motion control system with a double-acting cylinder. Other notable aspect of the formulation is the inclusion of cylinder friction and deadband in a proportional valve in the same framework along with the first-time consideration of the variable pressure at the pump delivery achieved by a hydro-mechanical compensator. The proposed model is more versatile than existing feedforward models constrained either for double-rod cylinders or a specific type of single-rod cylinder with the ratio of the flow areas across its piston matched with the ratio of the peripheral lengths of the corresponding metering ports in the valve sleeve. Of course, the effect of oil compressibility is expected to be more pronounced in force tracking than in motion tracking. Also, in an ageing valve, more energy is expected to get drained with the leakage flow. Considering all these aspects, the developed model indeed concluded as very versatile—applicable for both motion and force tracking by double acting cylinders of both single and double-rod types, new to ageing control valves and pump operation in both fixed- and variable-displacement modes.

In order to establish the validity of the model, real-time experiments have been conducted with a PI feedback to take care of modelling simplifications and approximations in the proposed and its closest existing feedforward models. The P and I gains have been estimated by a prior stability analysis of the linearized error equation with a proposed extension of Routh criteria for a time-variant system. Another real-time comparison has been provided against an existing nonlinear adaptive sliding-mode controller executable with constant pump pressure. With respect to it, the average power saving yield of both the FFPI controllers with variable pump pressure has also been quantified. All the comparisons have clearly established the proposed model as acceptable and superior.

Appendix

In (4), using (10), (22), (24) and (26) for $\dot{y}\geq 0$ and (11), (23), (25) and (27) for $\dot{y}<0$ in the expressions of f_1 to f_5, the coefficients a_2, a_1, a_0 and a_{-1} in (15) are expressed by employing (42–47) as

$$a_2 = N_2/D - f_{5_{\ddot{y}}} \tag{94}$$

$$a_1 = N_1/D - f_{5_{\ddot{y}}} \tag{95}$$

$$a_0 = N_0/D - f_{5_y} + f_{5_V}k_P \tag{96}$$

$$a_{-1} = k_I f_{5_{p_p}}\{f_{1_{Q_p}}f_{5_V}/(1-f_{1_{Q_P}}f_{5_{p_p}}) + N_{-1}/D\} \tag{97}$$

where

$$N_2 = f_{5_{p_p}} f_{1_{Q_p}} \left[f_{2_{p_1}} \left\{ f_{1_{Q_p}} (f_{3_{p_p}} f_{3_{P_2}} f_{4_{\ddot{y}}} - f_{4_{p_p}} f_{2_{p_2}} f_{3\ddot{y}}) + \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{3_{\ddot{y}}} \right\} + \left\{ f_{1_{Q_p}} f_{2_{p1}} (f_{4_{p_p}} f_{3_{\ddot{y}}} - f_{3_{p_p}} f_{4_{\ddot{y}}}) + \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{4_{\ddot{y}}} \right\} \right]$$
$$+ \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \left[f_{5_{P_1}} \left\{ \begin{pmatrix} f_{1_{Q_p}} f_{2_{p_2}} (f_{3_{p_p}} f_{4_{\ddot{y}}} - \\ f_{4_{p_p}} f_{3_{\ddot{y}}} \end{pmatrix} \right\} + f_{3_{\ddot{y}}} \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) + f_{5_{p_2}} \left\{ f_{1_{Q_p}} f_{2_{p1}} (f_{4_{p_p}} f_3 - f_{3_{p_p}} f_{4_{\ddot{y}}}) + f_{4_{\ddot{y}}} \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \right\} \right] \tag{98}$$

$$N_1 = f_{5_{p_p}} f_{1_{Q_p}} \left[f_{2_{p_1}} \left\{ f_{1_{Q_p}} f_{2_{p_2}} (f_{4_{p_p}} f_{3_{\ddot{y}}} - f_{3_{p_p}} f_{4\ddot{y}}) - \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{3_{\ddot{y}}} \right\} + f_{2_{p1}} \begin{Bmatrix} f_{1_{Q_p}} f_{2_{p1}} (f_{4_{p_p}} f_{3_{\ddot{y}}} - f_{3_{p_p}} f_{4_{\ddot{y}}}) \\ -\left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{4_{\ddot{y}}} \end{Bmatrix} \right] \tag{99}$$
$$+ \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \left[f_{5_{p_1}} \left\{ f_{1_{Q_p}} f_{2_{p_2}} (f_{3_{p_p}} f_{4_{\ddot{y}}} - f_{4_{p_p}} f_{3_{\ddot{y}}}) + f_{3_{\ddot{y}}} \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \right\} + \left\{ f_{1_{Q_p}} f_{2_{p1}} (f_{4_{p_p}} f_{3_{\ddot{y}}} - f_{3_{p_p}} f_{4_{\ddot{y}}}) + f_{4_{\ddot{y}}} \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \right\} \right]$$

$$N_0 = f_{5_{p_p}} f_{1_{Q_p}} \left\{ \left(f_{2_{p_1}} f_{3_y} + f_{2_{p_2}} f_{4_y}\right) \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) - k_P f_{1_{Q_p}} f_{2_V} \left(f_{3_{p_p}} f_{2_{p_1}} + f_{4_{p_p}} f_{2_{p_2}}\right) \right\}$$
$$+ \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) \begin{bmatrix} f_{5_{p_1}} \left\{ f_{1_{Q_p}} f_{2_{p_2}} \left(f_{3_{p_p}} f_{4_y} - f_{4_{p_p}} f_{3_y}\right) + \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{3_y} - k_P f_{1_{Q_p}} f_{3_{p_p}} f_{2_V} \right\} \\ + f_{5_{p_2}} \left\{ f_{1_{Q_p}} f_{2_{p_1}} \left(f_{4_{p_p}} f_{4_y} - f_{3_{p_p}} f_{3_y}\right) + \left(1 - f_{1_{Q_p}} f_{2_{p_p}}\right) f_{3_y} - k_P f_{1_{Q_p}} f_{4_{p_p}} f_{2_V} \right\} \end{bmatrix} \tag{100}$$
$$+ k_P f_{5_{p_2}} f_{1_{Q_p}} f_{2_{p_1}} f_{3_{p_p}} \left(1 - f_{1_{Q_p}} f_{4_{p_p}} f_{2_V}\right)$$

$$N_{-1} = f_{1_{Q_p}} f_{2_V} [f_{1_{Q_p}} f_{2_{p2}} f_{4_{p_p}} + f_{1_{Q_p}} f_{2_{p1}} f_{3_{p_p}} - (1 - f_{1_{Q_p}} f_{2_{p_p}}) \{ f_{5_{p1}} f_{3_{p_p}} + f_{5_{p2}} f_{4_{p_p}} \}] \tag{101}$$

and

$$D = \{(1 - f_{1_{Q_p}} f_{2_{p_p}}) - f_{1_{Q_p}} f_{2_{p_2}} f_{4_{p_p}}\} \{(1 - f_{1_{Q_p}} f_{2_{p_p}}) - f_{1_{Q_p}} f_{2_{p_1}} f_{3_{p_p}}\} - f_{1_{Q_p}}^2 f_{3_{p_p}} f_{4_{p_p}} f_{2_{p_1}} f_{2_{p_2}} \tag{102}$$

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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